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Nested Counterfactual Identification from Arbitrary Surrogate Experiments

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Abstract

The *Ladder of Causation* describes three qualitatively different types of activities an agent may be interested in engaging in, namely, seeing (observational), doing (interventional), and imagining (counterfactual) (Pearl and Mackenzie, 2018). The inferential challenge imposed by the causal hierarchy is that data is collected by an agent observing or intervening in a system (layers 1 and 2), while its goal may be to understand what would have happened had it taken a different course of action, contrary to what factually ended up happening (layer 3). While there exists a solid understanding of the conditions under which cross-layer inferences are allowed from observations to interventions, the results are somewhat scarcer when targeting counterfactual quantities. In this paper, we study the identification of nested counterfactuals from an arbitrary combination of observations and experiments. Specifically, building on a more explicit definition of nested counterfactuals, we prove the counterfactual unnesting theorem (CUT), which allows one to map arbitrary nested counterfactuals to unnested ones. For instance, applications in mediation and fairness analysis usually evoke notions of direct, indirect, and spurious effects, which naturally require nesting. Second, we introduce a sufficient and necessary graphical condition for marginal counterfactual identification from an arbitrary combination of observational and experimental distributions. Lastly, we develop a complete algorithm for identifying marginal nested counterfactuals and a finite algebraic procedure for specified conditional cells. Negative decisions witness nonidentification in the positive model class permitting finite observed-domain enlargement; no efficiency guarantee is made for the conditional procedure.

1 Introduction

Counterfactuals provide the basis for notions pervasive throughout human affairs, such as credit assignment, blame and responsibility, and regret. One of the most powerful constructs in human reasoning —"what if?" questions— evokes hypothetical conditions usually contradicting the factual evidence. Judgment and understanding of critical situations found from medicine to psychology to business involve counterfactual reasoning, e.g.: "Joe received the treatment and died, would he be alive had he not received it?," "Had the candidate been male instead of female, would the decision from the admissions committee be more favorable?," or "Would the profit this quarter remain within 5% of its value had we increased the price by 2%?". By and large, counterfactuals are key ingredients that go in the construction of explanations about why things happened as they did [17, 19].

The structural interpretation of causality provides proper semantics for representing counterfactuals [17, Ch. 7]. Specifically, each structural causal model (SCM) $\mathcal{M}$ induces a collection of distributions related to the activities of seeing (called observational), doing (interventional), and imagining (counterfactual). The collection of these distributions is known as the *Ladder of causation* [19], and has also been called the *Pearl's Causal Hierarchy* (PCH, for short) [2]. The PCH is a containment

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hierarchy; each type of distribution can be put in increasingly refined layers: observational content goes in layer 1; experimental in layer 2; counterfactual in layer 3; see Fig. 1.

It is understood that if we have all the information in the world about layer 1, there are still questions about layers 2 and 3 that are unanswerable, or technically undetermined; further, if we have data from layers 1 and 2, there are still questions in the world about layer 3 that are underdetermined [17, 2].

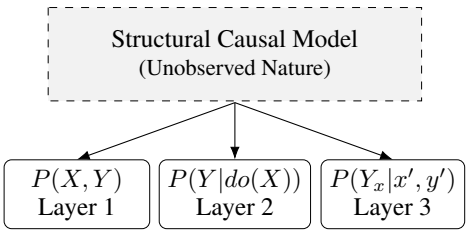


Figure 1: Every SCM induces different quantities in each layer of the PCH.

The inferential challenge in these settings arises because the generating model $\mathcal{M}$ is not fully observed, nor data from all of the layers are necessarily available, perhaps due to the cost or the infeasibility of performing certain interventions. One common task found in the literature is to determine the effect of an intervention of a variable X on an outcome Y, say $P(Y|do(X))$ (layer 2), using data from observations $P(\mathbf{V})$ (layer 1), where $\mathbf{V}$ is the set of observed variables, and possibly other interventions, e.g., $P(\mathbf{V}|do(Z))$. Also, qualitative assumptions about the system are usually articulated in the form of a causal diagram $\mathcal{G}$. This setting has been studied in the literature under the rubric of non-parametric identification from a combination of observations and experiments. Multiple solutions exist, including Pearl's celebrated do-calculus [16], and other increasingly refined methods that are computationally efficient, sufficient, and necessary [26, 9, 20, 27, 21, 11, 3, 14, 13].[1]

There is a growing literature about cross-layer inferences from data in layers 1 and 2 to quantities in layer 3. For example, a data scientist may be interested in evaluating the effect of an intervention on the group of subjects who receive the treatment instead of those randomly assigned to it. This measure is known as the *effect of treatment on the treated* [10, 17], and there exists a graphical condition for mapping it to a (layer 2) causal effect [24]. Further, there are also results on the identification of *path-specific effects*, which are counterfactuals that isolate specific paths in the graph [18, 1]. In particular, [25] provides a sufficient and necessary algorithm for identification of these effects from observational data, and [29] provides identification conditions from observational and experimental data in general canonical models. Further, [22] studied counterfactual identification under the assumption that all experimental distributions (i.e., over every subset of the observed variables) are available.[2]

In this paper, our goal is to identify the probability distribution of (possibly nested) counterfactual events from an *arbitrary* combination of user-specified observational and experimental distributions. To the best of our knowledge, this provides the first general treatment of nested counterfactual identification from arbitrary data collections. Moreover, it also provides the first, graphical and algorithmic, sufficient and necessary conditions for the identifications of counterfactuals from observational data alone (when no experimental data is available) and arbitrary causal diagrams. Moving up the PCH, our results allow for arbitrary quantities as inferential targets and for the addition of arbitrary experimental distributions to the input, increasing the flexibility of the solution.

For concreteness, consider the causal diagram shown in Fig. 2 and a counterfactual query called *direct effect*. This quantity represents the sensitivity of a variable Y to changes in another variable X while all other factors in the analysis remain fixed. Suppose X is level of exercise, M cholesterol levels, and Y cardiovascular disease. Exercising can improve cholesterol levels, which in turn affect the chances of developing cardiovascular disease. An interesting question is how much exercise prevents the disease by means other than regulating cholesterol. In counterfactual notation, this is to compare Y_{x,M_x} and Y_{x',M_x}, where x and x' are different values. The first quantity represents the value of Y when $X{=}x$ and M varies

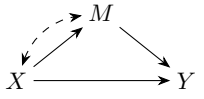


Figure 2: Causal diagram with treatment X, outcome Y, and mediator M.

[1] In fact, this is a classic task in a larger family of problems known as data fusion, which include other challenges such as selection bias, transportability, to cite a few. For more details, see [4].

[2] For the sake of context, the work proposed here can be seen as a generalization of two tasks, counterfactual identification under the assumptions discussed earlier [22] and interventional identification from arbitrary experiments [14]. As discussed later on, we will be able to show, based on the machinery developed here, that the individual methods for those tasks can be combined and also be shown complete.

accordingly. The second expression is the value Y attains if X is held constant at x' while M still follows $X{=}x$. The difference $E[Y_{x',M_x} - Y_{x,M_x}]$ — known as the natural direct effect (NDE) — measures the average direct effect under this comparison. In this instance, the nested counterfactual is identifiable from complete experiments on X; observational data alone are insufficient.

After all, there is no general identification method for this particular counterfactual family (which also includes indirect and spurious effects) and, more broadly, other arbitrary nested counterfactuals that are well-defined in layer 3. Our goal is to understand the non-parametric identification of arbitrary nested and conditional counterfactuals when the input consists of any combination of observational and interventional distributions, whatever is available for the data scientist. More specifically, our contributions are as follows.

1. We look at nested counterfactuals from an SCM perspective and introduce machinery that supports counterfactual reasoning. In particular, we prove the counterfactual unnesting theorem (CUT), which allows one to map any nested counterfactual to an unnested one (Section 2).
2. Building on this new machinery, we derive sufficient and necessary graphical conditions and an algorithm to determine the identifiability of marginal nested counterfactuals from an arbitrary combination of observational and experimental distributions (Section 3).
3. We reduce conditional identification to exact probability-ratio comparisons and derive a complete finite algebraic algorithm in the graph-uniform setting (Section 4).

Due to space constraints, all the proofs in this paper can be found in the full technical report [5].

1.1 Preliminaries

We denote variables by capital letters, X, and values by small letters, x. Bold letters, $\mathbf{X}$ represent sets of variables and $\mathbf{x}$ sets of values. The domain of a variable X is denoted by $\mathfrak{X}_X$. Two values $\mathbf{x}$ and $\mathbf{z}$ are said to be consistent if they share the common values for $\mathbf{X} \cap \mathbf{Z}$. We also denote by $\mathbf{x} \setminus \mathbf{Z}$ the value of $\mathbf{X} \setminus \mathbf{Z}$ consistent with $\mathbf{x}$ and by $\mathbf{x} \cap \mathbf{Z}$ the subset of $\mathbf{x}$ corresponding to variables in $\mathbf{Z}$. We assume the domain of every variable is finite.

We rely on causal graphs and denote them with a calligraphic letter, e.g., $\mathcal{G}$. We denote the set of vertices (i.e., variables) in $\mathcal{G}$ as $\mathbf{V}(\mathcal{G})$. Given a graph $\mathcal{G}$, $\mathcal{G}_{\overline{\mathbf{W}}\underline{\mathbf{X}}}$ is the result of removing edges coming into variables in $\mathbf{W}$ and going out from variables in $\mathbf{X}$. $\mathcal{G}[\mathbf{W}]$ denotes a vertex-induced subgraph including $\mathbf{W}$ and the edges among its elements. We use kinship notation for graphical relationships such as parents, children, descendants, and ancestors of a set of variables. For example, the set of parents of $\mathbf{X}$ in $\mathcal{G}$ is $Pa(\mathbf{X})_{\mathcal{G}} := \mathbf{X} \cup \bigcup_{X \in \mathbf{X}} Pa(X)_{\mathcal{G}}$. Similarly, we define $Ch()$, $De()$, and $An()$.

To articulate and formalize counterfactual questions, we require a framework that allows us to reason simultaneously about events from *alternative worlds*. Accordingly, we employ the Structural Causal Model (SCM) paradigm [17, Ch. 7]. An SCM $\mathcal{M}$ is a 4-tuple $\langle \mathbf{U}, \mathbf{V}, \mathcal{F}, P(\mathbf{u}) \rangle$, where $\mathbf{U}$ is a set of exogenous (latent) variables; $\mathbf{V}$ is a set of endogenous (observable) variables; $\mathcal{F}$ is a collection of functions such that each variable $V_i \in \mathbf{V}$ is determined by a function $f_i \in \mathcal{F}$. Each f_i is a mapping from a set of exogenous variables $\mathbf{U}_i \subseteq \mathbf{U}$ and a set of endogenous variables $\mathbf{Pa}_i \subseteq \mathbf{V} \setminus \{V_i\}$ to the domain of V_i. Uncertainty is encoded through a probability distribution over the exogenous variables, $P(\mathbf{U})$. An SCM $\mathcal{M}$ induces a *causal diagram* $\mathcal{G}$ where $\mathbf{V}$ is the set of vertices, there is a directed edge $(V_j \to V_i)$ for every $V_i \in \mathbf{V}$ and $V_j \in \mathbf{Pa}_i$, and a bidirected edge $(V_i \dashleftarrow\dashrightarrow V_j)$ for every pair $V_i, V_j \in \mathbf{V}$ such that $U_i \cap U_j \neq \emptyset$ (V_i and V_j have a common exogenous parent).

Distinct exogenous roots are mutually independent; shared exogenous parents represent the dependence encoded by bidirected edges. Independent private roots are allowed. We assume that the underlying model is recursive. That is, there are no cyclic dependencies among the variables. Equivalently, that is to say, that the corresponding causal diagram is acyclic. The set $\mathbf{V}(\mathcal{G})$ can be partitioned into subsets called *c-components* [28] such that two variables belong to the same c-component if they are connected in $\mathcal{G}$ by a path made entirely of bidirected edges.

2 SCMs and Nested Counterfactuals

Intervening on a system represented by an SCM $\mathcal{M}$ results in a new model differing only on the mechanisms associated with the intervened variables [15, 6, 7]. If the intervention consists on fixing the value of a variable X to a constant $x \in \mathfrak{X}_X$, it induces a *submodel*, denoted $\mathcal{M}_x$ [17, Def. 7.1.2].

To formally study nested counterfactuals, we extend this notion to models derived from interventions that replace functions from the original SCM with other, not necessarily constant, functions.

Definition 1 (Derived Model)**.** Let $\mathcal{M}$ be an SCM, $\widehat{\mathbf{U}} \subseteq \mathbf{U}$, $X \in \mathbf{V}$, and $\widehat{X} : \widehat{\mathbf{U}} \to \mathfrak{X}_X$ a function. Then, $\mathcal{M}_{\widehat{X}}$, called the *derived model* of $\mathcal{M}$ according to $\widehat{X}$, is identical to $\mathcal{M}$, except that the function f_X is replaced with a function $\widehat{f}_X$ identical to $\widehat{X}$.

This definition is easily extendable to models derived from an intervention on a set $\mathbf{X}$ instead of a singleton. When $\widehat{\mathbf{X}}$ is a collection of functions $\{\widehat{X} : \widehat{\mathbf{U}}_X \to \mathfrak{X}_X\}_{X \in \mathbf{X}}$, the derived model $\mathcal{M}_{\widehat{\mathbf{X}}}$ is obtained by replacing each f_X with $\widehat{X}$ for $X \in \mathbf{X}$. Next, we discuss the concept of *potential response* [17, Def. 7.4.1] with respect to the derived models.

Definition 2 (Potential Response)**.** Let $\mathbf{X}, \mathbf{Y} \subseteq \mathbf{V}$ be subsets of observable variables, let $\mathbf{u}$ be a unit, and let $\widehat{\mathbf{X}}(\mathbf{u})$ be a set of functions from $\widehat{\mathbf{U}}_X \to \mathfrak{X}_X$, for $X \in \mathbf{X}$ where $\widehat{\mathbf{U}}_X \subseteq \mathbf{U}$. Then, $\mathbf{Y}_{\mathbf{X}=\widehat{\mathbf{X}}}(\mathbf{u})$ (or $\mathbf{Y}_{\widehat{\mathbf{X}}}(\mathbf{u})$, for short) is called the *potential response* of $\mathbf{Y}$ to $\mathbf{X} = \widehat{\mathbf{X}}$, and is defined as the solution of $\mathbf{Y}$, for a particular $\mathbf{u}$, in the derived model $\mathcal{M}_{\widehat{\mathbf{X}}}$.

A potential response $Y_{\widehat{\mathbf{X}}}(\mathbf{u})$ describes the value that variable Y would attain for a unit (or individual) $\mathbf{u}$ if the intervention $\widehat{\mathbf{X}}$ is performed. This concept is tightly related to that of *potential outcome*, but the former explicitly allows for interventions that do not necessarily fix the variables in $\mathbf{X}$ to a constant value. Averaging over the space of $\mathbf{U}$, a potential response $Y_{\widehat{\mathbf{X}}}(\mathbf{u})$ induces a random variable that we will denote simply as $Y_{\widehat{\mathbf{X}}}$. If the intervention replaces a function f_X with a potential response of X in $\mathcal{M}$, we say the intervention is *natural*.

When variables are enumerated as $W_1, W_2, \ldots$, we may add square brackets around the part of the subscript denoting interventions. We use $\mathbf{W}_*$ to denote sets of arbitrary counterfactual variables. Let $\mathbf{W}_* = \{W_{1[\widehat{\mathbf{T}}_1]}, W_{2[\widehat{\mathbf{T}}_2]}, \ldots\}$ represent a set of counterfactual variables such that $W_i \in \mathbf{V}$ and $\mathbf{T}_i \subseteq \mathbf{V}$ for $i = 1, \ldots, l$. Define $\mathbf{V}(\mathbf{W}_*) = \{W \in \mathbf{V} \mid W_{\widehat{\mathbf{T}}} \in \mathbf{W}_*\}$, that is, the set of observables that appear in $\mathbf{W}_*$. Let $\mathbf{w}_*$ represent a vector of values, one for each variable in $\mathbf{W}_*$ and define $\mathbf{w}_*(\mathbf{X}_*)$ as the subset of $\mathbf{w}_*$ corresponding to $\mathbf{X}_*$ for any $\mathbf{X}_* \subseteq \mathbf{W}_*$.

The probability of any counterfactual event is given by

$$P(\mathbf{Y}_* = \mathbf{y}_*) = \sum\nolimits_{\{\mathbf{u} \mid \mathbf{Y}_*(\mathbf{u}) = \mathbf{y}_*\}} P(\mathbf{u}), \tag{1}$$

where the predicate $\mathbf{Y}_*(\mathbf{u}) = \mathbf{y}_*$ means $\bigwedge_{\{Y_{\widehat{\mathbf{X}}} \in \mathbf{Y}_*\}} Y_{\widehat{\mathbf{X}}}(\mathbf{u}) = y$.

When all variables in the expression have the same subscript, that is, they belong to the same submodel; we will often denote it as $P_{\mathbf{x}}(W_1, W_2, \ldots)$.

For most real-world scenarios, having access to a fully specified SCM of the underlying system is unfeasible. Nevertheless, our analysis does not rely on such privileged access but the aspects of the model captured by the causal graph and data samples generated by the unobserved model.

2.1 Nested Counterfactuals

Potential responses can be compounded based on natural interventions. For instance, the counterfactual $Y_{Z_x}(\mathbf{u})$ ($Y_{Z=Z_x}(\mathbf{u})$) can be seen as the potential response of Y to an intervention that makes $\widehat{Z}$ equal to Z_x. Notice that $Z_x(\mathbf{u})$ is in itself a potential response, but from a different (nested) model. Hence we call Y_{Z_x} a *nested counterfactual*.

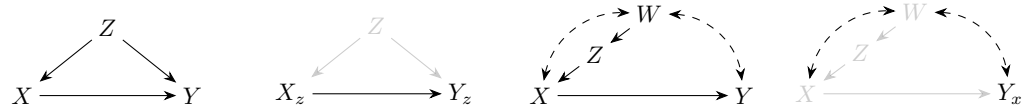
Recall the causal diagram in Fig. 2 and consider once again the NDE as

$$\mathit{NDE}_{x \to x', Z}(Y) = E[Y_{x' Z_x}] - E[Y_x]. \tag{2}$$

The second term is also equal to $Y_{x Z_x}$ as Z_x is consistent with $X = x$, so it is the value Y listens to in $\mathcal{M}_x$. Meanwhile, the first one is related to $P(Y_{x' Z_x})$, the probability of a nested counterfactual.

2.2 Tools for Counterfactual Reasoning

Before characterizing the identification of counterfactuals from observational and experimental data, we develop from first principles a canonical representation of any such query. First, we extend the notion of ancestors for counterfactual variables, which subsumes the usual one described before.



(a) "Backdoor" graph. (b) Graphical representation of the ancestors of Y_z. (c) "Napkin" graph. (d) Graphical representation of the ancestors of Y_x.

Figure 3: Two causal diagrams and the subgraphs considered when finding sets of ancestors for a counterfactual variable.

Definition 3 (Ancestors, of a counterfactual)**.** Let $Y_{\mathbf{x}}$ be a nonconstant response with $Y \in \mathbf{V} \setminus \mathbf{X}$ and $\mathbf{X} \subseteq \mathbf{V}$, after applying effectiveness to fixed responses. Then, the set of (counterfactual) ancestors of $Y_{\mathbf{x}}$, denoted $An(Y_{\mathbf{x}})$, consist of each $W_{\mathbf{z}}$, such that $W \in An(Y)_{\mathcal{G}_{\underline{\mathbf{X}}}}$ (which includes Y itself), and $\mathbf{z} = \mathbf{x} \cap An(W)_{\mathcal{G}_{\overline{\mathbf{X}}}}$.

For a set of variables $\mathbf{W}_*$, we define $An(\mathbf{W}_*)$ as the union of the ancestors of each variable in the set. That is, $An(\mathbf{W}_*) = \bigcup_{W_{\mathbf{t}} \in \mathbf{W}_*} An(W_{\mathbf{t}})$. For finite nested expressions the same definition is extended by expanding the structural equations. To evaluate a variable V in a world with intervention map I, return its assigned constant if it is fixed. If it is naturally assigned a response of V, expand that response in its own defining world. Otherwise, create an occurrence of f_V and recursively expand each observed parent in world I. Each occurrence retains its complete original parent slots and exogenous inputs; all occurrences of a primitive use the same function and exogenous variables. Intervention aliases are removed, and hard constants are slots rather than natural ancestor nodes. The resulting finite graph defines $An(\mathbf{Y}_*)$ for a nested query. Expansion preserves the response pointwise, and every functional edge follows an original parent–child edge in $\mathcal{G}$, so no occurrence of a primitive can be an ancestor of another occurrence of that primitive. The identification algorithm below takes this finite hard/natural response syntax as input; the unrestricted latent functions allowed in Definition 1 need not have such an input representation.

For instance, in Fig. 3(a), $An(Y_x) = \{Y_x, Z\}$, $An(X_{yz}) = \{X_z\}$ and $An(Y_z) = \{Y_z, X_z\}$ (depicted in Fig. 3(b)). In Fig. 3(c) $An(Z, Y_z) = \{Y_z, X_z, Z, W\}$ and $An(Y_x) = \{Y_x\}$ (represented in Fig. 3(d)).

Probabilistic and causal inference with graphical models exploits the local structure among variables, specifically parent-child relationships, to infer and even estimate probabilities. In particular, Tian [28] introduced *c-factors* which have proven instrumental in solving many problems in causal inference. We naturally generalize this notion to the counterfactual setting with the following definition.

Definition 4 (Counterfactual Factor (ctf-factor))**.** A counterfactual factor is a distribution of the form

$$P(W_{1[\mathbf{pa}_1]} = w_1, W_{2[\mathbf{pa}_2]} = w_2, \ldots, W_{l[\mathbf{pa}_l]} = w_l), \tag{3}$$

where each $W_i \in \mathbf{V}$ and there could be $W_i = W_j$ for some $i, j \in \{1, \ldots, l\}$.

For example, for Fig. 3(c) $P(Y_x = y, Y_{x'} = y')$, $P(Y_x = y, X_z = x)$ are ctf-factors but $P(Y_z = y, Z_w = z)$ is not. Using the notion of ancestrality introduced in Definition 3, we can factorize counterfactual probabilities as ctf-factors.

Theorem 1 (Ancestral set factorization)**.** *Let $\mathbf{W}_*$ be an ancestral set, that is, $An(\mathbf{W}_*) = \mathbf{W}_*$, and let $\mathbf{w}_*$ be a vector with a value for each variable in $\mathbf{W}_*$. Then,*

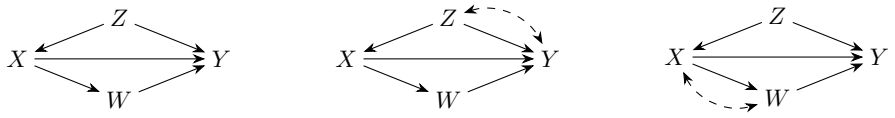
$$P(\mathbf{W}_* = \mathbf{w}_*) = P\left(\bigwedge_{W_{\mathbf{t}} \in \mathbf{W}_*} W_{\mathbf{pa}_w} = w\right), \tag{4}$$

where each w is w_t and $\mathbf{pa}_w$ is determined for each $W_{\mathbf{t}} \in \mathbf{W}_*$ as follows:

(i) the values for variables in $\mathbf{Pa}_w \cap \mathbf{T}$ are the same as in $\mathbf{t}$, and

(ii) the values for variables in $\mathbf{Pa}_w \setminus \mathbf{T}$ are taken from $\mathbf{w}_*$ corresponding to the parents of W.

Proof outline. Following a reverse topological order in $\mathcal{G}$, look at each $W_{i[\mathbf{t}_i]} \in \mathbf{W}_*$. Since any parent of W_i not in $\mathbf{T}_i$ must appear in $\mathbf{W}_*$, the composition axiom [17, 7.3.1] licenses adding them to the subscript. Then, by exclusion restrictions [16], any intervention not involving $Pa(W_i)$ can be removed to obtain the form in Eq. (4). □



(a) (b) (c)

Figure 4: Three causal diagrams representing plausible structures in mediation analysis.

For example, consider the diagram in Fig. 3(c) and the counterfactual $P(Y_x = y \mid X = x')$ known as the *effect of the treatment on the treated* (ETT) [10, 17]. First note that $P(Y_x = y \mid X = x') = P(Y_x = y, X = x')/P(X = x')$ and that $An(Y_x, X) = \{Y_x, X, Z, W\}$, then

$$P(Y_x = y, X = x') = \sum\nolimits_{z,w} P(Y_x = y, X = x', Z = z, W = w). \tag{5}$$

Then, by Theorem 1 we can write

$$P(Y_x = y, X = x') = \sum\nolimits_{z,w} P(Y_x = y, X_z = x', Z_w = z, W = w). \tag{6}$$

Moreover, the following result describes a factorization of ctf-factors based on the c-component structure of the graph, which will prove instrumental in the next section.

Theorem 2 (Counterfactual factorization)**.** *Let $P(\mathbf{W}_* = \mathbf{w}_*)$ be a ctf-factor, let $W_1 < W_2 < \cdots$ be a topological order over the variables in $\mathcal{G}[\mathbf{V}(\mathbf{W}_*)]$, and let $\mathbf{C}_1, \ldots, \mathbf{C}_k$ be the c-components of the same graph. Define $\mathbf{C}_{j*} = \{W_{\mathbf{pa}_w} \in \mathbf{W}_* \mid W \in \mathbf{C}_j\}$ and $\mathbf{c}_{j*}$ as the values in $\mathbf{w}_*$ corresponding to $\mathbf{C}_{j*}$, then $P(\mathbf{W}_* = \mathbf{w}_*)$ decomposes as*

$$P(\mathbf{W}_* = \mathbf{w}_*) = \prod\nolimits_j P(\mathbf{C}_{j*} = \mathbf{c}_{j*}). \tag{7}$$

Furthermore, each factor can be computed from the joint counterfactual table*, with every intervention value in its subscripts held fixed. Write $R_0 = 1$ and $R_i = P(\bigwedge_{\{W_{\mathbf{pa}_w} \in \mathbf{W}_* : W \le W_i\}} W_{\mathbf{pa}_w} = w)$. Whenever the displayed denominators are positive,*

$$P(\mathbf{C}_{j*} = \mathbf{c}_{j*}) = \prod_{\{i : W_i \in \mathbf{C}_j\}} \frac{R_i}{R_{i-1}}. \tag{8}$$

The probabilities R_i are marginals of that fixed counterfactual table; parent-completing values must not be varied while its output coordinates are marginalized.

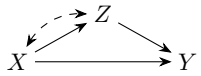
Armed with these results, we consider the identification problem in the next section.

3 Counterfactual Identification from Observations and Experiments

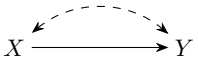
In this section, we consider the identification of a counterfactual probability from a collection of observational and experimental distributions. This task can be seen as a generalization of that in [14] where the available data is the same, but the query is a causal effect $P_{\mathbf{x}}(\mathbf{Y})$. Let $\mathbb{Z} = \{\mathbf{Z}_1, \mathbf{Z}_2, \ldots\}, \mathbf{Z}_j \subseteq \mathbf{V}$, and assume that all of $\{P_{\mathbf{z}_j}(\mathbf{V})\}_{\mathbf{z}_j \in \mathfrak{X}_{\mathbf{Z}_j}, \mathbf{Z}_j \in \mathbb{Z}}$ are available. Notice that $\mathbf{Z}_j = \emptyset$ is a valid choice corresponding to $P(\mathbf{V})$ the observational (non-interventional) distribution.

For the identification results in this section, we assume a positive observational law $P(\mathbf{V})$, whether or not it is supplied; the input contains every intervention value for each specified target. Identification is uniform over finite observed domains containing the named constants. Non-identifiability constructions may use a common finite enlargement of those domains in the two models, preserving all named values and their equalities. Thus completeness here does not assert completeness for every separately prescribed, fixed-cardinality model class.

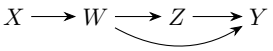
Definition 5 (Counterfactual Identification)**.** A query $P(\mathbf{Y}_* = \mathbf{y}_*)$ is said to be identifiable from $\mathbb{Z}$ in $\mathcal{G}$, if $P(\mathbf{Y}_* = \mathbf{y}_*)$ is uniquely computable from the distributions $\{P_{\mathbf{z}_j}(\mathbf{V})\}_{\mathbf{z}_j \in \mathfrak{X}_{\mathbf{Z}_j}, \mathbf{Z}_j \in \mathbb{Z}}$ in any causal model which induces $\mathcal{G}$.



(a) A complete experiment intervening on X and leaving Z unmanipulated identifies $P(Z_x = z)$.



(b) For $x \neq x'$, the factor $P(Y_x = y, X = x')$ is inconsistent.



(c) For $x \neq x'$, the factor $P(W_x = w, W_{x'} = w')$ is inconsistent.

Figure 5: Examples of causal diagrams and inconsistent ctf-factors derived from them.

Given an arbitrary query $P(\mathbf{Y}_* = \mathbf{y}_*)$, we could express it in terms of ctf-factors by writing $P(\mathbf{Y}_* = \mathbf{y}_*) = \sum_{\mathbf{d}_* \setminus \mathbf{y}_*} P(\mathbf{D}_* = \mathbf{d}_*)$ where $\mathbf{D}_* = An(\mathbf{Y}_*)$ and then using Theorem 1 to write $P(\mathbf{D}_* = \mathbf{d}_*)$ as a ctf-factor. For instance, the ancestral set $\mathbf{W}_* = \{Y_x, X, Z, W\}$ with $\mathbf{w} = \{y, x', z, w\}$ in Eq. (6) can be written in terms of ctf-factors as

$$P(Y_x = y, X_z = x', Z_w = z, W = w) = P(Y_x = y, X_z = x', W = w)P(Z_w = z). \quad (9)$$

The following lemma characterizes the relationship between the identifiability of $P(\mathbf{Y}_* = \mathbf{y}_*)$ and $P(\mathbf{D}_* = \mathbf{d}_*)$.

Lemma 1. *Let $\mathbf{W}_* = An(\mathbf{Y}_*)$, before rewriting its complete cells in parent-completed form. Then $P(\mathbf{Y}_* = \mathbf{y}_*)$ is identifiable from $\mathbb{Z}$ if and only if every complete cell $P(\mathbf{W}_* = \mathbf{w}_*)$ extending $\mathbf{Y}_* = \mathbf{y}_*$ is identifiable from $\mathbb{Z}$.*

Once the query of interest is in ctf-factor-form, the identification task reduces to identifying smaller ctf-factors according to the c-components of $\mathcal{G}$. In this respect, Theorem 2 implies the following.

Corollary 1. *Let $P(\mathbf{W}_*)$ be a joint counterfactual table with fixed intervention subscripts, and let $\mathbf{C}_{j*} \subseteq \mathbf{W}_*$ be one of its c-component subsets. If the marginal table $P(\mathbf{C}_{j*})$ is not identifiable, then the joint table $P(\mathbf{W}_*)$ is not identifiable.*

Proof. Identification of the joint table would identify each of its marginals. This is a statement about tables, not a converse from one factor cell to one joint cell. The latter step in the completeness proof below uses a localized countermodel and positive unchanged factors. □

Let us consider the causal diagrams in Fig. 4 and the counterfactual $Y_{x_1, W_{x_0}} = y, X = x$, with $x_0, x_1, x \in \mathfrak{X}_X$, used to define quantities for fairness analysis in [29] (e.g., $Y_{x_1, W_{x_0}} = y | X = x$):

$$\begin{aligned}
&P(Y_{x_1, W_{x_0}} = y, X = x) \\
&\quad = \sum\nolimits_{w} P(Y_{x_1, w} = y, W_{x_0} = w, X = x) && \text{Unnesting} && (10)\\
&\quad = \sum\nolimits_{w,z} P(Y_{x_1, w} = y, W_{x_0} = w, X = x, Z = z) && \text{Complete ancestral set} && (11)\\
&\quad = \sum\nolimits_{w,z} P(Y_{x_1, w, z} = y, W_{x_0} = w, X_z = x, Z = z) && \text{Write in ctf-factor-form} && (12)
\end{aligned}$$

Due to the particular c-component structure of each model, we can factorize $P(Y_{x_1,w,z} = y, W_{x_0} = w, X_z = x, Z = z)$ according to each model as:

$$P(Y_{x_1,w,z} = y)P(W_{x_0} = w)P(X_z = x)P(Z = z), \quad (13)$$
$$P(Y_{x_1,w,z} = y, Z = z)P(W_{x_0} = w)P(X_z = x), \text{ and} \quad (14)$$
$$P(Y_{x_1,w,z} = y)P(W_{x_0} = w, X_z = x)P(Z = z). \quad (15)$$

The question then becomes, whether ctf-factors corresponding to individual c-components can be identified from the available input. In this example, all factors in Eq. (13) and Eq. (14) are identifiable from $P(\mathbf{V})$. For Eq. (14) in particular, they are given by

$$P(Y = y \mid Z = z, W = w, X = x_1)P(Z = z)P(W = w \mid X = x_0)P(X = x \mid Z = z). \quad (16)$$

In contrast, the factor $P(W_{x_0}{=}w, X_z{=}x)$ in Eq. (15) (model Fig. 4(c)) is only identifiable if $x{=}x_0$. The following definition and theorem characterize the factors that can be identified from $\mathbb{Z}$ and $\mathcal{G}$.

Definition 6 (Inconsistent ctf-factor)**.** $P(\mathbf{W}_* = \mathbf{w}_*)$ is an inconsistent ctf-factor if it is a ctf-factor, $\mathcal{G}[\mathbf{V}(\mathbf{W}_*)]$ has a single c-component, and one of the following situations hold:

(i) there exist $W_{\mathbf{t}} \in \mathbf{W}_*$ and $Z \in \mathbf{T} \cap \mathbf{V}(\mathbf{W}_*)$ such that $z \in \mathbf{t}$, $z' \in \mathbf{w}_*$ and $z \neq z'$, or
(ii) there exist $W_{i[\mathbf{t}_i]}, W_{j[\mathbf{t}_j]} \in \mathbf{W}_*$ and $T \in \mathbf{T}_i \cap \mathbf{T}_j$ such that $t \in \mathbf{t}_i, t' \in \mathbf{t}_j$ and $t \neq t'$.

Theorem 3 (Ctf-factor identifiability)**.** *Let $P(\mathbf{W}_* = \mathbf{w}_*)$ be a consistent ctf-factor whose underlying variables form one c-component. Identical controlled responses are merged; incompatible outcomes on an identical response instead give probability zero. Put $\mathbf{W} = \mathbf{V}(\mathbf{W}_*)$ and $\mathbf{W}' = \mathbf{V} \setminus \mathbf{W}$. Then $P(\mathbf{W}_* = \mathbf{w}_*) = P_{\mathbf{w}'}(\mathbf{w})$, where $\mathbf{w}, \mathbf{w}'$ agree with all the outcomes and parent settings in the factor. Hence identification of this controlled kernel from $\mathbb{Z}$ identifies the factor. Consistency alone does not assert availability of that kernel from an arbitrary $\mathbb{Z}$.*

Consider the $NDE_{x \to x', Z}(Y)$ in Fig. 5(a), we can write

$$P(Y_{x'Z_x} = y) = \sum\nolimits_z P(Y_{x'z} = y, Z_x = z) = \sum\nolimits_z P(Y_{x'z} = y) P(Z_x = z). \tag{17}$$

While the factor $P(Y_{x'z} = y)$ is identifiable from $P(\mathbf{V})$ as $P(Y = y \mid X = x', Z = z)$, the second factor is identifiable only if experimental data on X is available, as $P_x(z)$.

We can also verify that the factor $P(Y_x = y, X = x')$ in Fig. 5(b) is inconsistent. For another example consider the ETT-like expression $P(Y_{x,z} = y, X = x', Z = z')$ in Fig. 5(c), we have

$$\begin{aligned}
&P(Y_{xz} = y, X = x', Z = z') \\
&\quad = \sum\nolimits_{w,w'} P(Y_{xz} = y, X = x', Z = z', W_x = w, W = w') && (18)\\
&\quad = \sum\nolimits_{w,w'} P(Y_{xz} = y, X = x', Z_{w'} = z', W_x = w, W_{x'} = w') && (19)\\
&\quad = \sum\nolimits_{w,w'} P(Y_{zw} = y) P(X = x') P(Z_{w'} = z') P(W_x = w, W_{x'} = w'), && (20)
\end{aligned}$$

where the factor $P(W_x = w, W_{x'} = w')$ is inconsistent.

Given a counterfactual variable $\mathbf{Y}_{\mathbf{x}}$, it could be the case that some values in $\mathbf{x}$ become causally irrelevant to Y after the rest of $\mathbf{x}$ has been fixed. Formally,

Lemma 2 (Interventional Minimization)**.** *Let $Y_{\mathbf{x}}$ be a counterfactual variable, $\mathcal{G}$ a causal diagram, and*

$$Y_{\mathbf{z}} \text{ such that } \mathbf{Z} = \mathbf{X} \cap An(Y)_{\mathcal{G}_{\overline{\mathbf{X}}}} \text{ and } \mathbf{z} = \mathbf{x} \cap \mathbf{Z}. \tag{21}$$

Then, $Y_{\mathbf{z}} = Y_{\mathbf{x}}$ holds for any model compatible with $\mathcal{G}$. Moreover, this transformation is denoted as $\|Y_{\mathbf{x}}\| := Y_{\mathbf{z}}$ and $\|\mathbf{Y}_*\| := \bigcup_{Y_{\widehat{\mathbf{X}}} \in \mathbf{Y}_*} \|Y_{\widehat{\mathbf{X}}}\|$.

Moreover, such simplification may reveal counterfactual expressions with equivalent or contradicting events. In Fig. 3(c), $(Y_{xz} = y, Y_{xz'} = y') = (Y_x = y, Y_x = y')$ which has probability 0 if $y \neq y'$, or $(Y_{xz} = y, Y_{xz'} = y)$ that is simply $(Y_x = y)$. Similarly, the probabilities of counterfactual events of the form $P(X_x = x')$, $x \neq x'$, and $P(X_x = x)$ are trivially 0 and 1 respectively.

The same minimization applies to natural interventions: retain the replacement functions only for targets in $\mathbf{X} \cap An(Y)_{\mathcal{G}_{\overline{\mathbf{X}}}}$. The proof in the supplement already evaluates these functions pointwise. For a vector outcome take the union of these ancestor sets. Apply this minimization before unnesting, including within inner natural responses.

The following result shows how nested counterfactuals can be written in terms of non-nested ones.

Theorem 4 (Counterfactual Unnesting Theorem (CUT))**.** *Let $\widehat{\mathbf{X}}, \widehat{\mathbf{Z}}$ be any natural interventions on disjoint sets $\mathbf{X}, \mathbf{Z} \subseteq \mathbf{V}$. Then, for $\mathbf{Y} \subseteq \mathbf{V}$ disjoint from $\mathbf{X}$ and $\mathbf{Z}$ such that $\mathbf{X} \subseteq An(\mathbf{Y})_{\mathcal{G}_{\overline{\mathbf{X} \cup \mathbf{Z}}}}$, $P(\mathbf{Y}_{\widehat{\mathbf{Z}}, \widehat{\mathbf{X}}} = \mathbf{y})$ is identifiable iff $P(\mathbf{Y}_{\widehat{\mathbf{Z}}, \mathbf{x}} = \mathbf{y}, \widehat{\mathbf{X}} = \mathbf{x})$ is identifiable for every $\mathbf{x}$, and given by*

$$P(\mathbf{Y}_{\widehat{\mathbf{Z}}, \widehat{\mathbf{X}}} = \mathbf{y}) = \sum\nolimits_{\mathbf{x} \in \mathfrak{X}_{\mathbf{X}}} P(\mathbf{Y}_{\widehat{\mathbf{Z}}, \mathbf{x}} = \mathbf{y}, \widehat{\mathbf{X}} = \mathbf{x}). \tag{22}$$

For instance, for the model in Fig. 2 we can write

$$P(Y_{x'Z_x} = y) = \sum\nolimits_z P(Y_{x'z} = y, Z_x = z). \tag{23}$$

Algorithm 1 CTFIDU($\mathbf{Y}_*, \mathbf{y}_*, \mathbb{Z}, \mathcal{G}$)

Input: $\mathcal{G}$ causal diagram over variables $\mathbf{V}$; $\mathbf{Y}_*$ a set of counterfactual variables in $\mathbf{V}$; $\mathbf{y}_*$ a set of values for $\mathbf{Y}_*$; and available distribution specification $\mathbb{Z}$.
Output: $P(\mathbf{Y}_* = \mathbf{y}_*)$ in terms of available distributions or FAIL if not identifiable from $\langle \mathcal{G}, \mathbb{Z} \rangle$.

let $\mathbf{Y}_* \leftarrow \|\mathbf{Y}_*\|$.
if there exists $Y_{\mathbf{x}} \in \mathbf{Y}_*$ with two or more different values in $\mathbf{y}_*(Y_{\mathbf{x}})$ or $Y_y \in \mathbf{Y}_*$ with $\mathbf{y}_*(Y_y) \neq y$ **then return** 0.
if there exists $Y_{\mathbf{x}} \in \mathbf{Y}_*$ with two consistent values in $\mathbf{y}_*(Y_{\mathbf{x}})$ or $Y_y \in \mathbf{Y}_*$ with $\mathbf{y}_*(Y_y) = y$ **then** remove repeated variables from $\mathbf{Y}_*$ and values $\mathbf{y}_*$.
Expand the finite structural response expressions; return 0 if a queried constant conflicts with its outcome, remove true constant events, and return 1 if the query is empty.
Apply the value-aware equality reduction described above, obtaining $(\mathcal{G}', \mathbf{Y}'_*, \mathbf{y}'_*)$; return 0 on incompatible merged outcomes.
let $(\mathbf{Y}_*, \mathbf{y}_*) \leftarrow (\mathbf{Y}'_*, \mathbf{y}'_*)$ and let $\mathbf{W}_*$ be the natural nodes of $\mathcal{G}'$, with their inherited original parent slots; form the ctf-factors $\mathbf{C}_{1*}, \ldots, \mathbf{C}_{k*}$ for the c-components $\mathbf{C}_i$ of $\mathcal{G}[\mathbf{V}(\mathbf{W}_*)]$.
for each $\mathbf{C}_i$ s.t. $(\mathbf{C}_{i*} = \mathbf{c}_{i*})$ is not inconsistent, $\mathbf{Z} \in \mathbb{Z}$ s.t. $\mathbf{C}_i \cap \mathbf{Z} = \emptyset$ **do**
  let $\mathbf{B}_i$ be the c-component of $\mathcal{G}[\mathbf{V} \setminus \mathbf{Z}]$ containing $\mathbf{C}_i$; compute $P_{\mathbf{V}\setminus\mathbf{B}_i}(\mathbf{B}_i)$ from the supplied kernel $P_{\mathbf{Z}}(\mathbf{V} \setminus \mathbf{Z})$.
  if IDENTIFY$(\mathbf{C}_i, \mathbf{B}_i, P_{\mathbf{V}\setminus\mathbf{B}_i}(\mathbf{B}_i), \mathcal{G})$ does not FAIL **then**
    let $P_{\mathbf{V}\setminus\mathbf{C}_i}(\mathbf{C}_i) \leftarrow$ IDENTIFY$(\mathbf{C}_i, \mathbf{B}_i, P_{\mathbf{V}\setminus\mathbf{B}_i}(\mathbf{B}_i), \mathcal{G})$.
    let $P(\mathbf{C}_{i*} = \mathbf{c}_{i*}) \leftarrow P_{\mathbf{V}\setminus\mathbf{C}_i}(\mathbf{C}_i)$ evaluated with values $(\mathbf{c}_{i*} \cup \bigcup_{C_{\mathbf{t}} \in \mathbf{C}_{i*}} \mathbf{pa}_c)$.
    move to the next $\mathbf{C}_i$.
  end if
end for
if any $P(\mathbf{C}_{i*} = \mathbf{c}_{i*})$ is inconsistent or was not identified from $\mathbb{Z}$ **then return** FAIL.
return $P(\mathbf{Y}_* = \mathbf{y}_*) \leftarrow \sum_{\mathbf{w}_* \setminus \mathbf{y}_*} \prod_i P(\mathbf{C}_{i*} = \mathbf{c}_{i*})$.

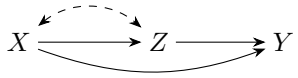
For general finite nested conjunctions, the algorithm expands the structural expressions jointly and normalizes their shared equations before forming factors. The unnesting identity remains available, but non-identifiability is not inferred independently for arbitrary summands of that identity.

Using the results in this section, we propose CTFIDU (Algorithm 1) for a finite counterfactual conjunction built from hard interventions and finite natural-response expressions. First expand the nested expressions into their structural ancestor graph as defined above. A natural response is evaluated in its own defining world. Reduce constant query outputs by effectiveness; an empty remaining conjunction has probability one. After intervention minimization, the procedure also identifies responses that must be equal on the requested event. In primitive topological order, merge two natural copies of the same variable whenever their corresponding original parent slots are identical or have equal values fixed by the query or interventions; transfer query assignments and return zero if they conflict. Repeat until no such merge remains, preserving one complete original parent-slot map at each retained node. These are the value-aware equality operations of Shpitser and Pearl [23, Lemmas 24–25]. They preserve the query event for each exogenous realization. Let $\mathcal{G}'$ be the resulting graph restricted to natural ancestors of the transformed query. The graph is used for equality reduction; each node still evaluates its original equation and primitive noise.

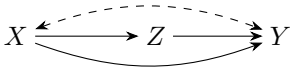
The relevant ctf-factors are formed from the natural nodes of $\mathcal{G}'$. For the consistency test, use one free value symbol for each unqueried natural node and the named constants for queried or intervened values. Settings must agree as symbols throughout a factor, not merely for selected summation values. Equivalently, assign fresh, distinct labels to unqueried nodes before testing. This prevents rejection on an impossible raw summand. For example, on $X \to Z \to Y$, the event $\{X = 0, Y_{X=0} = 0, Y = 0\}$ reduces to $\{X = 0, Y = 0\}$; the two Z copies must merge before their descendants are tested.

For a consistent factor the existing IDENTIFY subroutine obtains $P_{\mathbf{V}\setminus\mathbf{C}_i}(\mathbf{C}_i)$ from an available source whenever possible. The source loop is complete by Kivva et al. [12, Theorem 1]. The procedure returns FAIL if a normalized factor is inconsistent or its controlled kernel cannot be recovered from any supplied source. Otherwise it returns the original sum of products.

Theorem 5 (CTFIDU completeness)**.** *In the positive, domain-flexible graph-uniform class specified above, a finite counterfactual probability, built from hard interventions and finite natural-response*



(a) While $P(Y_x{=}y \mid Z_x{=}z, X{=}x')$ is identifiable from $\mathbb{Z}$, $P(Y_x{=}y, Z_x{=}z, X{=}x')$ is not.



(b) $P(Y_x{=}y \mid Z_x{=}z, X{=}x')$ is not graph-uniformly identifiable, even from all hard experiments.

Figure 6: Examples of conditional queries, with $x \neq x'$ and observational data available.

expressions, $P(\mathbf{Y}_* = \mathbf{y}_*)$ is identifiable from $\mathbb{Z}$ and $\mathcal{G}$ if and only if CTFIDU *returns an expression for it.*

This result ascertains that Algorithm 1 solves the identification task for counterfactuals in the form $P(\mathbf{Y}_* = \mathbf{y}_*)$.[3] Still, there exist other quantities that take into account evidence from events observed in the system, and cannot be written in this particular form. Since these queries represent important aspects of the underlying system, we will discuss them in the next section.

4 Identification of Conditional Counterfactuals

In this section, we consider counterfactual quantities of the form $P(\mathbf{Y}_* = \mathbf{y}_* \mid \mathbf{X}_* = \mathbf{x}_*)$. It is immediate to write such a query as $P(\mathbf{W}_* = \mathbf{w}_*)/\sum_{\mathbf{y}_*} P(\mathbf{W}_* = \mathbf{w}_*)$ with $\mathbf{W}_* = \mathbf{Y}_* \cup \mathbf{X}_*$, and try to identify it using CTFIDU. Nevertheless, depending on the graphical structure, the original query may be identifiable even if the latter is not. To witness, consider the causal diagram in Fig. 6(a) and the counterfactual $P(Y_x = y \mid Z_x = z, X = x')$, which can be written as $P(Y_x = y, Z_x = z, X = x')/\sum_y P(Y_x = y, Z_x = z, X = x')$. For $x \neq x'$, and with observational data among the supplied inputs, the numerator is equal to $P(Y_{xz} = y)P(Z_x = z, X = x')$, where the second ctf-factor is inconsistent, and therefore not identifiable from $\mathbb{Z}$. Nevertheless, the conditional query is identifiable as

$$\frac{P(Y_{xz} = y)P(Z_x = z, X = x')}{P(Z_x = z, X = x')\sum_y P(Y_{xz} = y)} = P(Y_{xz} = y) = P(Y = y \mid Z = z, X = x). \tag{24}$$

For Fig. 6(b), take independent uniform roots $U \in \{0, 1, 2\}$, $R \in \{0, 1, 2, 3\}$, and $W \in \{0, 1\}$. Set $X = U$, $Z = W$, and $Y = \mathbf{1}\{R < k(X, U)\}$. In two models let $k(0, \cdot)$ equal $(2, 1, 3)$ and $(2, 3, 1)$, respectively, with all other entries equal to 2. Both models have a positive observational distribution and agree on every complete hard-intervention kernel: the natural diagonal is unchanged, and each fixed-X row has the same sum. Yet $P(Y_0 = 1 \mid Z_0 = 0, X = 1)$ is $1/4$ and $3/4$, respectively. This witnesses nonidentification in the graph-uniform finite-domain class; it does not assert nonidentification under a binary-X restriction.

To characterize such simplifications, it is necessary to retain the entire conditional ratio. An outcome-specific equality can simplify its numerator without simplifying its evidence event. For example, on $X \to Y$, the event $X = x$ permits replacing Y by Y_x in a joint numerator; that replacement cannot be carried into a denominator in which $X = x$ is no longer imposed. We therefore form the two events separately, keeping every named value fixed:

$$A = \{\mathbf{Y}_* = \mathbf{y}_*\}, \qquad B = \{\mathbf{X}_* = \mathbf{x}_*\},$$
$$P(A \mid B) = \frac{N}{D}, \qquad N = P(A \cap B), \quad D = P(B) > 0. \tag{25}$$

Finite natural nesting is evaluated by recursive substitution in each event, or expanded separately using Theorem 4.

The following algebraic reduction gives a complete treatment of cancellation. Its purpose is to decide the specified cell, rather than require identification of a larger joint table. For finite observed alphabets, let R_V denote the full response table of primitive V. Connected cumulants of functions of these tables provide coordinates θ in which N, D are polynomials. All complete hard-intervention kernels

[3] The normalized counterfactual reduction of Shpitser and Pearl [23] can also be combined with the corrected general identification result of Kivva et al. [12]; see Section F. Practitioners familiar with these algorithms may find this equivalent approach useful.

Algorithm 2 CTFID$(\mathbf{Y}_*, \mathbf{y}_*, \mathbf{X}_*, \mathbf{x}_*, \mathbb{Z}, \mathcal{G})$

Input: a causal diagram, a finite hard/natural conditional query, and complete supplied experimental kernels as specified by $\mathbb{Z}$.

Output: an identifying expression on the positive-evidence domain; FAIL for graph-uniform nonidentification; or UNDEFINED if the evidence is impossible on every admissible finite domain.

1: Form $N = P(A \cap B)$ and $D = P(B)$ separately. Preserve the named equality pattern.
2: If separate CTFIDU calls identify both probabilities and the evidence is possible, return their ratio on $D > 0$.
3: **for** each admissible singleton-domain pattern **do**
4: Substitute its constants; recompute the residual expression graphs, m, and c_V. If $m = 0$ or no primitive remains, evaluate the constant events directly.
5: On a nonempty residual graph, apply the bounded full-experiment test at $c_V + 3m + 1$ states. Skip patterns with identically impossible evidence. If the ratio test fails, return FAIL.
6: For each remaining nonconstant pattern, construct the source-coordinate comparison and test its finite alphabet grid. If an identity fails, return FAIL.
7: **end for**
8: If every pattern has impossible evidence, return UNDEFINED.
9: **return** the reduced source-coordinate ratio of Lemma 3, constructed for the actual input alphabets and their singleton pattern, on the original positive-evidence domain.

together, denoted P^*, are equivalent to a linear map $L\theta$. Choose independent visible coordinates $s = L_0\theta$ and complementary coordinates h. Appendix D constructs these objects and their realizable open neighborhood.

Lemma 3 (Conditional Query Reduction)**.** *On fixed finite observed alphabets, write $N = g\bar{N}$ and $D = g\bar{D}$ in the polynomial coordinates (s, h), where g is a greatest common divisor and $\bar{N}, \bar{D}$ are coprime. The specified cell is identified from P^* if and only if both reduced polynomials are independent of h. On success their ratio is defined at every compatible input for which the original evidence has positive probability.*

For arbitrary supplied complete experiments, express that ratio in full intrinsic corner coordinates q, and again cancel the polynomial greatest common divisor. The cell is identified if and only if both resulting polynomials depend only on the corner blocks recoverable from at least one supplied experiment.

The corner coordinates in this lemma are the conditional parameters of the positive nested Markov parameterization [8]. They record the conditional distribution of the head of a connected set given its other vertices, after intervention on its external parents. A source identifies precisely its intrinsic blocks; their union gives the available coordinates. The proof also constructs actual graph-compatible countermodels with positive original evidence whenever an unavailable coordinate remains. It does not infer a conditional counterexample from a change in its numerator alone.

A fixed-alphabet test would not by itself terminate a decision over unrestricted finite alphabets. Two finite bounds resolve this issue. Let m count all structural-function evaluations needed to evaluate $A \cap B$ and B, including recursively evaluated ancestors and natural assignments, and let c_V count the distinct values of V named in the query. A full-experiment failure always has a witness with

$$|\mathcal{X}_V| \leq c_V + 3m + 1. \tag{26}$$

After full-experiment success, a finite polynomial comparison in the corner tables determines whether the ratio uses unavailable source coordinates. Its syntax gives a finite grid of alphabet sizes on which polynomial identity testing is sufficient. Singleton domains are covered by finitely many exact constant substitutions. Appendix D proves these bounds and gives every construction used below.

Theorem 6 (CTFID completeness)**.** *In the positive, domain-flexible graph-uniform class, a specified conditional cell with finite hard/natural response syntax is identifiable from $\mathbb{Z}$ and $\mathcal{G}$ if and only if the revised* CTFID *returns an expression for it. The statement concerns its positive-evidence domain; universally impossible evidence is reported separately. Every negative decision has two finite graph-compatible models agreeing on all supplied complete kernels and disagreeing on the specified cell, with positive observational laws and positive evidence in both.*

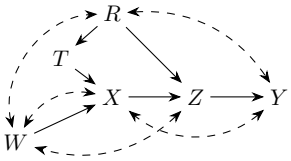


Figure 7: Causal diagram for the conditional running example.

The algebraic procedure is finite but can be extremely expensive. The marginal identification routine remains an immediate sufficient shortcut. The revised conditional algorithm uses the full ratio test. The earlier ancestral-component split can remove relevant evidence, as the example below shows, and its success or failure is not used as an identification decision.

Consider the diagram in Fig. 7 and the query

$$P(Y_{x_1 Z_{x_0}} = y_1 \mid X_{w_0} = x_1, T_{r_1} = t_1, R = r_0), \qquad r_0 \neq r_1, \tag{27}$$

with observational data and complete experiments on $\{W, T\}$. On the event $R = r_0$, the natural T entering X_{w_0} is T_{r_0}. This response and the evidence variable T_{r_1} share the same structural mechanism and private noise. Their joint factor $P(T_{r_0} = t, T_{r_1} = t_1)$ must therefore be retained. It cannot be replaced by $P(T_{r_0} = t)$ or canceled as an independent evidence factor.

In fact, Eq. (27) is not identified even from all complete hard-intervention kernels. The positive binary models in Section E agree on every such kernel, while the specified conditional cell is $3/4$ in one model and $1/4$ in the other. This example illustrates why shared primitive responses must be kept together and why identification must be checked for the original ratio.

The same diagram also gives an identified comparison. If the evidence uses $T_{r_0} = t_1$, consistency fixes the natural T to t_1, and

$$\begin{aligned} &P(Y_{x_1 Z_{x_0}} = y_1 \mid X_{w_0} = x_1, T_{r_0} = t_1, R = r_0) \\ &\quad = \sum_z P_{w_0, t_1}(y_1 \mid r_0, x_1, z) P_{w', t'}(z \mid r_0, x_0), \end{aligned} \tag{28}$$

where w' and t' are arbitrary fixed values. Both factors are available from the complete experiments on $\{W, T\}$.

5 Conclusions

We investigated in this paper the problem of nested and non-nested counterfactual identification from an arbitrary combination of observational and experimental distributions. First, we introduced fundamental building blocks for counterfactual reasoning, which allowed us to prove several properties of nested counterfactual distributions, including the counterfactual unnesting theorem (Theorem 4), and the ancestral and counterfactual factorization theorems (Theorem 1, Theorem 2). Moreover, we introduced a graphical condition (Definition 6, Theorem 3) and developed an efficient algorithm (Algorithm 1) for identifying marginal counterfactuals. We then proved their sufficiency and necessity for the task of nested counterfactual identification (Theorem 5). Lastly, we treated conditional identification through reduction of the entire probability ratio (Lemma 3) and provided a complete algebraic procedure (Algorithm 2 and Theorem 6). Its finite alphabet bounds establish termination in the graph-uniform setting; they do not establish an efficient conditional algorithm. These results advance the state of the art regarding the distributions the inference engine expects as input, and the query it can generate as the output. In terms of the input, it accepts any combination of observational and experimental distributions, and in terms of the output, it considers an arbitrary nested counterfactual distribution. This work closes the long-standing inferential gap within the layers of the Pearl's Causal Hierarchy, and now one can move within all layers of the hierarchy in a very flexible and general way.

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Nested Counterfactual Identification from Arbitrary Surrogate Experiments

Supplemental Material

A Counterfactual Unnesting Theorem

Theorem 4 (Counterfactual Unnesting Theorem (CUT))**.** *Let* $\widehat{\mathbf{X}}, \widehat{\mathbf{Z}}$ *be any natural interventions on disjoint sets* $\mathbf{X}, \mathbf{Z} \subseteq \mathbf{V}$*. Then, for* $\mathbf{Y} \subseteq \mathbf{V}$ *disjoint from* $\mathbf{X}$ *and* $\mathbf{Z}$ *such that* $\mathbf{X} \subseteq An(\mathbf{Y})_{\mathcal{G}_{\overline{\mathbf{X}\cup\mathbf{Z}}}}$*,* $P(\mathbf{Y}_{\widehat{\mathbf{Z}},\widehat{\mathbf{X}}} = \mathbf{y})$ *is identifiable iff* $P(\mathbf{Y}_{\widehat{\mathbf{Z}},\mathbf{x}} = \mathbf{y}, \widehat{\mathbf{X}} = \mathbf{x})$ *is identifiable for every* $\mathbf{x}$*, and given by*

$$P(\mathbf{Y}_{\widehat{\mathbf{Z}},\widehat{\mathbf{X}}} = \mathbf{y}) = \sum\nolimits_{\mathbf{x}\in\mathfrak{X}_{\mathbf{X}}} P(\mathbf{Y}_{\widehat{\mathbf{Z}},\mathbf{x}} = \mathbf{y}, \widehat{\mathbf{X}} = \mathbf{x}). \tag{22}$$

Proof. (soundness) From Eq. (1) we have: $P(\mathbf{Y}_{\widehat{\mathbf{Z}},\widehat{\mathbf{X}}} = \mathbf{y}) = \sum_{\{\mathbf{u}|\mathbf{Y}_{\widehat{\mathbf{Z}},\widehat{\mathbf{X}}}(\mathbf{u})=\mathbf{y}\}} P(\mathbf{u})$. In $\mathcal{M}_{\widehat{\mathbf{Z}}\widehat{\mathbf{X}}}$, for any given $\mathbf{u}$, the variables in $\mathbf{X}$ take a particular value-vector $\mathbf{x} = \widehat{\mathbf{X}}(\mathbf{u})$. Then, we can partition $\{\mathbf{u} \mid \mathbf{Y}_{\widehat{\mathbf{Z}},\widehat{\mathbf{X}}}(\mathbf{u}) = \mathbf{y}\}$ as $\bigcup_{\mathbf{x}\in\mathfrak{X}_{\mathbf{X}}}\{\mathbf{u} \mid \mathbf{Y}_{\widehat{\mathbf{Z}},\mathbf{X}=\mathbf{x}}(\mathbf{u}) = \mathbf{y}, \widehat{\mathbf{X}}(\mathbf{u}) = \mathbf{x}\}$. Hence

$$P(\mathbf{Y}_{\widehat{\mathbf{Z}},\widehat{\mathbf{X}}} = \mathbf{y}) = \sum_{\mathbf{x}\in\mathfrak{X}_{\mathbf{X}}} \sum_{\{\mathbf{u}|\mathbf{Y}_{\widehat{\mathbf{Z}},\mathbf{x}}(\mathbf{u})=\mathbf{y}, \widehat{\mathbf{X}}(\mathbf{u})=\mathbf{x}\}} P(\mathbf{u}) \quad = \sum_{\mathbf{x}\in\mathfrak{X}_{\mathbf{X}}} P(\mathbf{Y}_{\widehat{\mathbf{Z}},\mathbf{x}} = \mathbf{y}, \widehat{\mathbf{X}} = \mathbf{x}). \tag{29}$$

(necessity) Suppose that, for some $\mathbf{x}$, two models $\mathcal{M}^{(1)}$ and $\mathcal{M}^{(2)}$ agree on every supplied complete experimental distribution but disagree on

$$q_i = P^{(i)}(\mathbf{Y}_{\widehat{\mathbf{Z}},\mathbf{x}} = \mathbf{y}, \widehat{\mathbf{X}} = \mathbf{x}) = P^{(i)}(\mathbf{Y}_{\widehat{\mathbf{Z}},\widehat{\mathbf{X}}} = \mathbf{y}, \widehat{\mathbf{X}} = \mathbf{x}).$$

For each $X_j \in \mathbf{X}$, choose a directed path from X_j to an outcome in $\mathbf{Y}$ whose vertices after X_j avoid $\mathbf{X} \cup \mathbf{Z}$. Such paths exist by the stated ancestry condition. Enlarge the finite domain of each vertex D after a path's starting vertex with a coordinate $T_D^j \in \mathfrak{X}_{X_j}$. On the first edge set T_D^j equal to the old component of X_j; on each later edge copy the predecessor's jth coordinate. Use separate coordinates when paths intersect. The old structural functions continue to use only the old components of their parents. These operations use only edges already in $\mathcal{G}$, and no variable in $\mathbf{X} \cup \mathbf{Z}$ is enlarged.

Under any hard intervention, projection onto the old components gives the original intervened model. Conditional on all old values and the complete enlarged intervention values, the added coordinates are determined by the same recursive map in both models. Every supplied enlarged experimental distribution is thus a common pushforward of the corresponding old distribution. This includes every enlarged intervention value, not just values used in the query.

Use one injective embedding of the old domain of each variable into its enlarged domain for all occurrences of its values, including constants in inner interventions. Induction through each finite natural-response expression shows that its old component is unchanged, because the old structural functions ignore the added coordinates. In particular, $\widehat{\mathbf{X}}$ and $\widehat{\mathbf{Z}}$ retain their original values. In the outer derived model each selected path is uninterrupted, so every endpoint records $\widehat{X}_j$. Choose the enlarged value of each outcome to have old component y and coordinate x_j on every path through it. Then

$$\{\mathbf{Y}'_{\widehat{\mathbf{Z}},\widehat{\mathbf{X}}} = \mathbf{y}'\} = \{\mathbf{Y}_{\widehat{\mathbf{Z}},\widehat{\mathbf{X}}} = \mathbf{y}, \widehat{\mathbf{X}} = \mathbf{x}\}.$$

There is only one requested value per outcome variable in this theorem, so its injection can map y to precisely this y'. Rename the embedded states with their old labels and give other states fresh labels. This preserves the query's literal constants and all their equalities, even when an outcome value is also used as an inner clamp. The two enlarged models agree on the supplied inputs and give probabilities $q_1 \neq q_2$ for the original nested query.

For the positive model class, replace each deterministic recording channel by a common channel that copies with probability $1-\epsilon$ and otherwise chooses a uniform record. An independent private exogenous variable implements each channel and is shared across worlds. Every supplied experimental law is the old law multiplied by the same recording-channel factors. It is positive when the old law is positive. The query contrast is a polynomial in ϵ, nonzero at zero, and hence nonzero for sufficiently small positive ϵ. This proves necessity in the stated class with unrestricted finite domains.

□

B Proofs on ctf-factors and factorization results

Lemma 2 (Interventional Minimization)**.** *Let $Y_{\mathbf{x}}$ be a counterfactual variable, $\mathcal{G}$ a causal diagram, and*

$$Y_{\mathbf{z}} \text{ such that } \mathbf{Z} = \mathbf{X} \cap An(Y)_{\mathcal{G}_{\overline{\mathbf{X}}}} \text{ and } \mathbf{z} = \mathbf{x} \cap \mathbf{Z}. \tag{21}$$

Then, $Y_{\mathbf{z}} = Y_{\mathbf{x}}$ holds for any model compatible with $\mathcal{G}$. Moreover, this transformation is denoted as $\|Y_{\mathbf{x}}\| := Y_{\mathbf{z}}$ and $\|\mathbf{Y}_*\| := \bigcup_{Y_{\widehat{\mathbf{x}}} \in \mathbf{Y}_*} \|Y_{\widehat{\mathbf{X}}}\|$.

Proof. Let us fix a $\mathbf{u}$, so a counterfactual variable $Y_{\widehat{\mathbf{X}}}$ takes the value of the potential response $Y_{\widehat{\mathbf{X}}}(\mathbf{u})$. Note that this potential response is equal to $f_y((\mathbf{Pa}_y)_{\widehat{\mathbf{X}}}(\mathbf{u}), \mathbf{u})$. Next, for each $W \in \mathbf{Pa}_y$, we consider the response $W_{\widehat{\mathbf{X}}}(\mathbf{u})$. If $W \in \mathbf{X}$, $W_{\widehat{\mathbf{X}}}(\mathbf{u}) = \widehat{W}(\mathbf{u})$ where $\widehat{W} \in \widehat{\mathbf{X}}$. Else, we can evaluate $W_{\widehat{\mathbf{X}}}(\mathbf{u})$ as $f_w((\mathbf{Pa}_w)_{\widehat{\mathbf{X}}}(\mathbf{u}), \mathbf{u})$. Then, by following this procedure recursively, we evaluate every function in $An(Y)_{\mathcal{G}_{\underline{\mathbf{X}}}}$ and use the variables/values of $\widehat{\mathbf{X}}$ only for those variables in $\mathbf{Z} = \mathbf{X} \cap Pa\big(An(Y)_{\mathcal{G}_{\underline{\mathbf{X}}}}\big)_{\mathcal{G}} = \mathbf{X} \cap An(Y)_{\mathcal{G}_{\overline{\mathbf{X}}}}$.

It follows that evaluating $Y_{\widehat{\mathbf{Z}}}(\mathbf{u})$, for $\widehat{\mathbf{X}} \cap \mathbf{Z}$ and the same $\mathbf{u}$ will result in exactly the same value as $Y_{\widehat{\mathbf{X}}}(\mathbf{u})$, and the result follows. If this equality holds for any $\mathbf{u}$, it will hold for any averaging over them as well. □

Theorem 1 (Ancestral set factorization)**.** *Let $\mathbf{W}_*$ be an ancestral set, that is, $An(\mathbf{W}_*) = \mathbf{W}_*$, and let $\mathbf{w}_*$ be a vector with a value for each variable in $\mathbf{W}_*$. Then,*

$$P(\mathbf{W}_* = \mathbf{w}_*) = P\left(\textstyle\bigwedge_{W_{\mathbf{t}} \in \mathbf{W}_*} W_{\mathbf{pa}_w} = w\right), \tag{4}$$

where each w is w_t and $\mathbf{pa}_w$ is determined for each $W_{\mathbf{t}} \in \mathbf{W}_*$ as follows:

(i) the values for variables in $\mathbf{Pa}_w \cap \mathbf{T}$ are the same as in $\mathbf{t}$, and

(ii) the values for variables in $\mathbf{Pa}_w \setminus \mathbf{T}$ are taken from $\mathbf{w}_*$ corresponding to the parents of W.

Proof. Let $W_1 < W_2 < \ldots < W_k$ be a topological order of the variables in $\mathbf{W}$ according to $\mathcal{G}$. We transform the expression considering the instances of each W_i at a time, in reverse topological order. Let $W_{i\mathbf{t}_i}$ be any instance of W_i in $\mathbf{W}_*$. For every $W \in \mathbf{Pa}_{W_i} \setminus \mathbf{T}_i$, the variable $W_{\mathbf{t}}$ where and $\mathbf{T} = \mathbf{T}_i \cap An(W)_{\mathcal{G}_{\overline{\mathbf{T}_i}}}$ must be in $\mathbf{W}_*$ as $W < W_i$. Moreover, $W_{\mathbf{t}} = W_{\mathbf{t}_i}$ because interventions on $\mathbf{T}_i \setminus \mathbf{T}$ have no effect on $W_{\mathbf{t}}$. Then, using composition for W_i and all such parent W we get $W_{i[\mathbf{t}_i]} = W_{i[\mathbf{t}_i \cup (\mathbf{pa}_{w_i} \setminus \mathbf{t}_i)]}$, which is equal to $W_{i[\mathbf{pa}_{w_i}]}$. Repeating this process for $i = k, k-1, \ldots, 1$ ends with Eq. (4). □

Theorem 2 (Counterfactual factorization)**.** *Let $P(\mathbf{W}_* = \mathbf{w}_*)$ be a ctf-factor, let $W_1 < W_2 < \cdots$ be a topological order over the variables in $\mathcal{G}[\mathbf{V}(\mathbf{W}_*)]$, and let $\mathbf{C}_1, \ldots, \mathbf{C}_k$ be the c-components of the same graph. Define $\mathbf{C}_{j*} = \{W_{\mathbf{pa}_w} \in \mathbf{W}_* \mid W \in \mathbf{C}_j\}$ and $\mathbf{c}_{j*}$ as the values in $\mathbf{w}_*$ corresponding to $\mathbf{C}_{j*}$, then $P(\mathbf{W}_* = \mathbf{w}_*)$ decomposes as*

$$P(\mathbf{W}_* = \mathbf{w}_*) = \textstyle\prod_j P(\mathbf{C}_{j*} = \mathbf{c}_{j*}). \tag{7}$$

Furthermore, each factor can be computed from the joint counterfactual table*, with every intervention value in its subscripts held fixed. Write $R_0 = 1$ and $R_i = P(\bigwedge_{\{W_{\mathbf{pa}_w} \in \mathbf{W}_* : W \leq W_i\}} W_{\mathbf{pa}_w} = w)$. Whenever the displayed denominators are positive,*

$$P(\mathbf{C}_{j*} = \mathbf{c}_{j*}) = \prod_{\{i : W_i \in \mathbf{C}_j\}} \frac{R_i}{R_{i-1}}. \tag{8}$$

The probabilities R_i are marginals of that fixed counterfactual table; parent-completing values must not be varied while its output coordinates are marginalized.

Proof. Equation (7) follows from independence of the primitive root sets belonging to distinct c-components after every observed parent argument has been fixed. For extraction, use the fixed counterfactual table specified in the theorem, and let

$$H_{j,i} = P\left(\bigwedge_{\{W_{\mathbf{pa}_w} \in \mathbf{C}_{j*}: W \le W_i\}} W_{\mathbf{pa}_w} = w\right), \qquad H_{j,0} = 1.$$

The same independence gives $R_i = \prod_j H_{j,i}$. When $W_i \in \mathbf{C}_j$, all $H_{l,i}$ with $l \neq j$ equal $H_{l,i-1}$. Thus $R_i/R_{i-1} = H_{j,i}/H_{j,i-1}$ wherever the denominator is positive. Multiplication over all $W_i \in \mathbf{C}_j$ telescopes to $P(\mathbf{C}_{j*} = \mathbf{c}_{j*})$, establishing Eq. (8). Each R_i is obtained by marginalizing output coordinates of the fixed joint table; no intervention argument is changed by that marginalization. □

C Proofs of the results on Identification

Lemma 4. *Suppose $P(\mathbf{W}_* = \mathbf{w}_*)$ is not identifiable from the supplied complete experimental distributions in $\mathcal{G}$. Let $A_{1[\mathbf{t}_1]}, A_{2[\mathbf{t}_2]} \in \mathbf{W}_*$, where the latter response has the former as a natural ancestor. Then $\sum_{a_1} P(\mathbf{W}_* = \mathbf{w}_*)$ is not identifiable from the same inputs.*

Proof. The response expressions may contain finite, well-founded hard and natural interventions as specified in the recursive ancestor definition. Put $E = \{\mathbf{W}_* \setminus \{A_{1[\mathbf{t}_1]}\} = \mathbf{w}_* \setminus \{a_1\}\}$ and let H contain its natural ancestral response occurrences, before parent completion. Since the removed response is ancestral to a retained response, H contains $\mathbf{W}_*$. A pair of models witnessing the premise must differ on some complete H-cell h extending $\mathbf{w}_*$: the differences on these finitely many cells sum to the difference on $\mathbf{W}_* = \mathbf{w}_*$. We construct a pair that differs on E.

Apply the value-aware merges of Shpitser and Pearl [23, Lemmas 24–25] to the ancestral structural-expression graph until no permitted merge remains. Natural intervention aliases have already been expanded in their own defining worlds; each remaining edge is an original observed-parent input to a structural equation. Retain every original observed-parent slot, including fixed intervention values. A merge identifies copies of one primitive whose corresponding parents are identical or have equal values prescribed by the query or interventions. Transfer query assignments; conflicting assignments imply an empty event. Write K and E_K for the resulting graph and query.

Every natural ancestor retains a representative in K. To verify this, consider merging m into n. Their children become children of the representative. If an incoming edge $p \to m$ is lost, either its natural parent p also parents n, or p has a query-assigned value used to justify the merge. In the first case its path to a query continues through the representative; in the second, p is itself queried. Earlier ancestors keep their paths to p. Thus no surviving natural ancestor is removed by the final ancestor restriction. Moreover, on E each original occurrence equals its representative. Conversely, on E_K, prove this invariant by parents-first induction in the original graph. Corresponding natural parents already equal their representatives; query-assigned and fixed parents retain the values that justified each merge. Thus each original occurrence and its representative have the same parent tuple and output. Consequently a nonempty complete H-cell extending E corresponds to exactly one complete K-cell a extending E_K, and

$$\mathbf{1}\{H = h\} = \mathbf{1}\{K = a\}. \tag{30}$$

An assignment violating a merge equality instead denotes the empty event in every model. We distinguish two cases.

A primitive has several natural copies in K. Give each unqueried node a fresh selected value, distinct from all named values and other unqueried values of the same primitive; retain queried values. Also give each primitive V a further value $\perp_V$. Let r_n be the complete parent row selected at node n. Two remaining copies of the same primitive have distinct selected rows. Otherwise every corresponding parent equality would follow from identical nodes or query/intervention values, allowing another merge.

Choose a primitive V with $k \ge 2$ copies. Its single original structural equation returns the selected output a_i at row r_i when a private bit B_i is zero, and $\perp_V$ otherwise. At unselected rows it returns

$\perp_V$. Every other primitive returns the selected output on each of its selected rows and its $\perp$ value elsewhere. These prescriptions are unambiguous because the selected rows are distinct. In one model all B_i equal one fair bit B; in the other let $B_1 = B$, $B_2 = 1 - B$, and $B_i = B$ for $i > 2$. Every individual row bit is fair in both models. A hard-intervention regime reads only one row of this private vector, whose parents are determined before the vector is used. All other equations agree, so *every* complete hard-intervention kernel agrees, including kernels at the added states.

A copy returning $\perp_V$ causes every descendant on an uncut natural path to return its own $\perp$ value, since these values occur in no selected parent row. Every retained copy is ancestral to a query node. Hence E_K, and therefore the original E, holds exactly when $B_1 = \cdots = B_k = 0$. Its probabilities are $1/2$ and 0. For strictly positive kernels, add independent private switches that reset each primitive to a uniform state with probability ϵ, sharing each switch across its counterfactual copies. Every new hard kernel is a common mixture of old kernels under additional interventions, so equality of all hard kernels persists. All cells become positive; the query difference is polynomial in ϵ and remains nonzero for sufficiently small positive ϵ.

Every primitive has at most one natural copy in K. Use the original witnessing pair and its differing cell in Eq. (30). In a topological order of $\mathcal{G}$, enlarge each state space to

$$\mathcal{X}'_V = \mathcal{X}_V \times \mathcal{R}_V, \qquad \mathcal{R}_V = \prod_{U \in \mathrm{Pa}_{\mathcal{G}}(V)} \mathcal{X}'_U,$$

where $\mathrm{Pa}_{\mathcal{G}}(V)$ denotes the observed parents. For the unique natural V-node in K, let $p_V(a)$ be its selected complete parent row. Choose one injection $\iota_V : \mathcal{X}_V \to \mathcal{X}'_V$ satisfying

$$\iota_V(a_V) = \big(a_V, (\iota_U(p_V(a)_U))_{U \in \mathrm{Pa}_{\mathcal{G}}(V)}\big). \tag{31}$$

Map its other states, or all states if V is absent from K, to pairs with their original first component and any fixed second component. This is well defined in topological order and injective. Relabel each $\iota_V(c)$ by the original name c and give the other enlarged states fresh names. A value used both as an outcome and as an intervention therefore retains one common meaning.

In both models replace each original structural equation by

$$F'_V((x'_U)_U, u_V) = \big(F_V((\mathrm{pr}_1\, x'_U)_U, u_V), (x'_U)_U\big).$$

Thus the old equation ignores all new coordinates, while the added coordinate records the actual enlarged parent tuple. This is one original equation per primitive, shared across its worlds. The same normalization remains valid because the injections preserve every named equality. In the enlarged model the original event E equals exactly the selected base cell $K = a$. Indeed, a queried value in Eq. (31) forces its natural parents to have their prescribed enlarged values; repeating this reaches every natural ancestor. At a fixed input slot the full enlarged value is already clamped, so the argument stops there. Conversely, on the selected base cell, topological induction gives every node precisely its prescribed lift. The enlarged models therefore disagree on E.

For completeness, equality of the supplied distributions holds for every enlarged clamp assignment, not only the named values. For a complete cell v' under intervention t' on T, write r_V for the record part of v'_V, and include t' when evaluating parent values. Then

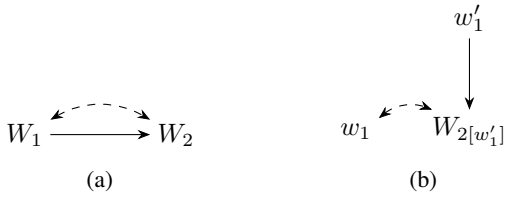
$$P'_{t'}(v'_{\mathbf{V}\setminus T}) = P_{\mathrm{pr}_1\, t'}(\mathrm{pr}_1\, v'_{\mathbf{V}\setminus T}) \prod_{V \notin T} \mathbf{1}\{r_V = (v'_U)_{U \in \mathrm{Pa}_{\mathcal{G}}(V)}\}. \tag{32}$$

The first factor agrees by the premise, and the remaining factors are common. To retain positivity, replace each record, independently across primitives, by a uniform record with probability δ. Use one private switch and draw per primitive, shared across all worlds. Each indicator in Eq. (32) becomes

$$(1 - \delta)\mathbf{1}\{r_V = (v'_U)_U\} + \delta/|\mathcal{R}_V|.$$

All required kernels remain equal and positive. The query difference is polynomial in δ and nonzero at zero, so it remains nonzero for sufficiently small positive δ. This construction uses only the supplied kernels; it does not require equality under additional intervention targets. Both cases prove the claim. □

Lemma 1. *Let $\mathbf{W}_* = An(\mathbf{Y}_*)$, before rewriting its complete cells in parent-completed form. Then $P(\mathbf{Y}_* = \mathbf{y}_*)$ is identifiable from $\mathbb{Z}$ if and only if every complete cell $P(\mathbf{W}_* = \mathbf{w}_*)$ extending $\mathbf{Y}_* = \mathbf{y}_*$ is identifiable from $\mathbb{Z}$.*



(a) (b)

Figure 8: (Case i, $l = 2$)

Proof. If every complete ancestral cell extending $\mathbf{Y}_* = \mathbf{y}_*$ is identified, its finite sum identifies $P(\mathbf{Y}_* = \mathbf{y}_*)$. Conversely, suppose one such cell is not identified. Every natural ancestor outside $\mathbf{Y}_*$ has a natural child on a path to a retained query response. Remove these ancestors in topological order and apply Lemma 4 at each step. Each construction preserves all named values and all supplied complete experimental kernels. The final event is the original $\mathbf{Y}_* = \mathbf{y}_*$, which is therefore not identifiable. Parent completion may rewrite each ancestral cell without changing its event, so the same conclusion applies to its factor representation. □

Lemma 5 (Inconsistent factor non-identifiability)**.** *Consider a complete selected cell of a normalized structural ancestor graph with one natural node per primitive, and its parent-completed conjunction. If a c-component has conflicting settings, that complete event is not identifiable from $\mathbf{P}_\star$ in the domain-flexible class.*

Proof. The complete event has one controlled response and one output per primitive. It is not identically zero: its prescribed rows can be implemented by the original equations. Its conflicting-settings component is therefore an instance of Shpitser and Pearl [23, Theorems 31(a)–32], applied to the parent-completed unnested conjunction. Their pair agrees on $\mathbf{P}_\star$ and differs on this complete event. Common private resets give positive hard kernels while preserving full-$\mathbf{P}_\star$ equality and a nonzero contrast for sufficiently small reset probability. □

The following constructions illustrate the conflicting-settings cases.

Proof. We construct two models $\mathcal{M}^{(1)}$ and $\mathcal{M}^{(2)}$ agreeing on $\mathbf{P}_\star^{(1)}$ and $\mathbf{P}_\star^{(2)}$ disagreeing on the inconsistent factor. Without loss of generality, let us ignore parents (intervened) that are consistent. Further, let $W_{1[\mathbf{t}_1]}$ and $W_{l[\mathbf{t}_l]}$ be inconsistent ones such that (i) $z = \mathbf{w}_*(W_{1[\mathbf{t}_1]})$ and $z' \in \mathbf{t}_l = \{z'\}$ are different and (ii) $t \in \mathbf{t}_1 = \{t\}$ and $t' \in \mathbf{t}_l = \{t'\}$ are different. We let $\mathbf{U} = \{U_1, \ldots, U_{l-1}\}$ be the unobserved variables where $W_i \leftarrow U_i \rightarrow W_{i+1}$ unless stated otherwise.

(Case i) Let $W_{1[\mathbf{t}_1]} = W_1$ and $W_{l[\mathbf{t}_l]} = W_{l[w_1']}$. We investigate the case with regard to $l = 2$ and $l > 2$.

(Case $l = 2$) Let U_1 be uniform on three states, let W_1 be ternary, and let $U = U_1$ for simplicity.

$$W_1(u) = u \tag{33}$$

$$W_l^{(1)}(w_1, u) = 1_{w_1=1,u=2} \qquad W_l^{(2)}(w_1, u) = 1_{w_1=1,u=0} \tag{34}$$

Then, two models agree on $\mathbf{P}_\star$. In the observational distribution, W_1 values are determined by the common values (i.e., $W_l^{(1)}$ and $W_l^{(2)}$ behave the same when $w_1 = u$). For interventional distributions, W_1 behaves the same marginally over U. However, on the event $W_1 = 0$ we have $U = 0$, while the W_1 input to $W_{2[w_1=1]}$ is intervened to 1. Thus $W_{2[w_1=1]}$ equals 0 in $\mathcal{M}^{(1)}$ and 1 in $\mathcal{M}^{(2)}$.

(Case $l > 2$) Let W_1 and U_1 be ternaries and others be binaries with $P(\mathbf{U})$ a uniform distribution. Let

$$\begin{aligned} W_1(u_1) &= u_1 \\ W_2(u_1, u_2) &= 1_{u_1=0,u_2=1} + 1_{u_1=2,u_2=0} \\ \forall_{3 \le i \le l-1} W_i(u_{i-1}, u_i) &= u_{i-1} \oplus u_i \\ W_l^{(1)}(w_1, u_{l-1}) &= 1_{w_1=1,u_{l-1}=1} \qquad W_l^{(2)} = 1_{w_1=1,u_{l-1}=0}. \end{aligned}$$

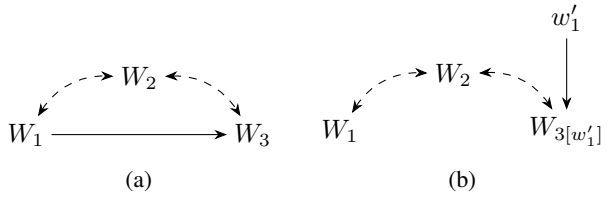


Figure 9: (Case i, $l > 1$)

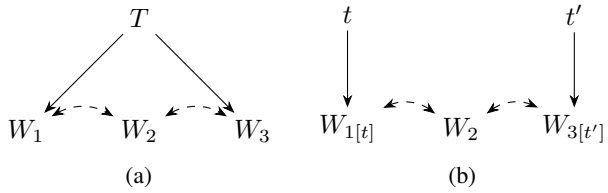


Figure 10

We first show that they agree on the observation for each U_1 value. Let $\mathbf{C}_{a:b} = \{C_a, \ldots, C_b\}$ for $a \leq b$ and $\emptyset$ otherwise.

(**Subcase** $U_1 = 0$) $W_1 = 0$ and $W_l = 0$. $\mathbf{W}_{2:l-1}$ behaves the same by construction.

(**Subcase** $U_1 = 1$) $W_1 = 1$ and $W_2 = 0$. Two models behave the same for $\mathbf{W}_{3:l}$ in a complementing $\mathbf{u}_{2:l-1}$:

$$\mathbf{W}^{(1)}_{3:l}(\mathbf{u}_{2:l-1}) = \mathbf{W}^{(2)}_{3:l}(\overline{\mathbf{u}_{2:l-1}}). \tag{35}$$

Combined with the uniformness of $P(\mathbf{U}_{2:l-1})$, $P(\mathbf{W}|U_1 = 1)$ are the same for both models.

(**Subcase** $U_1 = 2$) $W_1 = 2$ and $W_l = 0$ are fixed and $\mathbf{W}_{2:l-1}$ behave the same.

From the three subcases, both models agree on the observation.

Now we show they agree on $do(W_1 = w_1)$ (intervening on other variables is trivial, and intervening other variables together with W_1 becomes trivial once the two models agree on $do(W_1 = w_1)$.). In the case of $do(W_1 = 0)$ or $do(W_1 = 2)$, $W_l = 0$ and $\mathbf{W}_{2:l-1}$ behave the same. Hence, done. Now we focus on $do(W_1 = 1)$. We reuse a similar argument as shown in Equation (35). Let $\overline{u_1} = 2 - u_1$. Then, $(\mathbf{W}_{2:l})^{(1)}(\mathbf{u}) = (\mathbf{W}_{2:l})^{(2)}(\overline{\mathbf{u}})$. The common private-reset construction in Lemma 4 gives positive hard kernels while preserving their equality and, for sufficiently small reset probability, the counterfactual contrast.

Finally, we show that they disagree on $P(W_1 = 0, \mathbf{W}_{2:l-1}, W_{l,[w_1=1]})$. With $W_1 = 0$ observed, $U_1 = 0$. Then, $\bigoplus \mathbf{W} = \bigoplus \mathbf{W}_{2:l} = U_{l-1} \oplus W_l$, which is 0 for $\mathcal{M}^{(1)}$ and 1 for $\mathcal{M}^{(2)}$. Hence, they disagree.

(Case ii) Let T and the variables in $\mathbf{U}$ be independent fair coins. For $l = 1$, use the single equation $W^{(1)}(t, u) = u$ and $W^{(2)}(t, u) = u \oplus t$: the hard-intervention marginals agree, whereas $P(W_0 = 0, W_1 = 0)$ is $1/2$ and 0. For $l \geq 2$, let

$$\begin{aligned} W^{(1)}_1(u_1, t) &= u_1 & W^{(2)}_1(u_1, t) &= u_1 \oplus t \\ \forall_{2 \leq i \leq l-1} W_i(u_{i-1}, u_i) &= u_{i-1} \oplus u_i \\ W^{(1)}_l(u_{l-1}, t) &= u_{l-1} & W^{(2)}_l(u_{l-1}, t) &= u_{l-1} \oplus t. \end{aligned}$$

First, both models share $P(\mathbf{W}|t)$. For any t value and model, $\mathbf{U}$ maps to $\mathbf{W}$ uniformly where their bit-parity equals to zero. However, the bit-parity of $(W_{1[t=0]}, \mathbf{W}_{2:l-1}, W_{l[t=1]})$ differ for the models since T is counted only once in $\mathcal{M}^{(2)}$, and, the bit-parity becomes odd. □

Theorem 3 (Ctf-factor identifiability)**.** *Let $P(\mathbf{W}_* = \mathbf{w}_*)$ be a consistent ctf-factor whose underlying variables form one c-component. Identical controlled responses are merged; incompatible outcomes*

on an identical response instead give probability zero. Put $\mathbf{W} = \mathbf{V}(\mathbf{W}_*)$ *and* $\mathbf{W}' = \mathbf{V} \setminus \mathbf{W}$. *Then* $P(\mathbf{W}_* = \mathbf{w}_*) = P_{\mathbf{w}'}(\mathbf{w})$*, where* $\mathbf{w}, \mathbf{w}'$ *agree with all the outcomes and parent settings in the factor. Hence identification of this controlled kernel from* $\mathbb{Z}$ *identifies the factor. Consistency alone does not assert availability of that kernel from an arbitrary* $\mathbb{Z}$*.*

Proof. Let $\mathbf{z} = \mathbf{w}_* \cup \bigcup_{\{W_{\mathbf{pa}_w} \in \mathbf{W}_*\}} \mathbf{pa}_w$. Since the factor is consistent, $\mathbf{z}$ contains at most one value per observable. We can write $P(\mathbf{W}_* = \mathbf{w}) = P(\bigwedge_{\{W_{\mathbf{pa}_w} \in \mathbf{W}_*\}} W_{\mathbf{pa}_w} = w)$ where every $\mathbf{pa}_w$ is consistent with $\mathbf{z}$.

Let $W \in \mathbf{W}$ be such that $\mathbf{Pa}_w \cap \mathbf{W} = \emptyset$, then $\mathbf{Pa}_w \subseteq \mathbf{W}'$ and $W_{\mathbf{pa}_w} = W_{\mathbf{w}'}$ where $\mathbf{w}'$ has any valid value for $\mathbf{W}' \setminus \mathbf{Z}$. For a W with parents in $\mathbf{W}$, suppose every $T \in \mathbf{Pa}_w \setminus \mathbf{W}'$ already appears as $T_{\mathbf{w}'}$. On the co-conjunct event that these parent outputs equal their prescribed values, composition gives $W_{[\mathbf{pa}_W \cup \mathbf{w}']} = W_{\mathbf{w}'}$. Applying this event equality from parents to children in $\mathbf{W}$ we get

$$P(\mathbf{W}_* = \mathbf{w}) = P\Big(\bigwedge\nolimits_{\{W_{\mathbf{pa}_w} \in \mathbf{W}_*\}} W_{\mathbf{w}'} = w \Big) = P_{\mathbf{w}'}(\mathbf{w}). \tag{36}$$

The resulting equality is a controlled-kernel reduction. Recovering that kernel from $\mathbb{Z}$ is the separate causal identification step. A statement about one scalar cell is not obtained by invoking a table-recovery argument. □

C.1 Algorithm Completeness

Theorem 5 (CTFIDU completeness)**.** *In the positive, domain-flexible graph-uniform class specified above, a finite counterfactual probability, built from hard interventions and finite natural-response expressions,* $P(\mathbf{Y}_* = \mathbf{y}_*)$ *is identifiable from* $\mathbb{Z}$ *and* $\mathcal{G}$ *if and only if* CTFIDU *returns an expression for it.*

Proof. Soundness. Finite structural expansion, effectiveness, intervention minimization and the value-aware equality operations preserve the original requested event $E = \{\mathbf{Y}_* = \mathbf{y}_*\}$. Write K for the retained natural graph and E_K for its transformed query. Every natural node keeps its original primitive equation, root inputs and observed-parent slots. For a complete assignment a extending E_K, let $p_n(a)$ be the complete original parent row of node n, using the selected natural-parent value or the fixed intervention constant in each slot. If $v(n)$ is its primitive name, topological evaluation gives

$$\mathbf{1}\{K = a\} = \prod_{n \in K} \mathbf{1}\{f_{v(n)}(p_n(a), U_{v(n)}) = a_n\},$$

$$P(E) = \sum_{a \supseteq E_K} \prod_i H_i(a), \qquad H_i(a) = P\left(\bigwedge_{\{n : v(n) \in \mathbf{C}_i\}} f_{v(n)}(p_n(a), U_{v(n)}) = a_n \right). \tag{37}$$

The product uses independence of the primitive roots of distinct c-components. When the symbolic consistency test passes, each factor is the controlled kernel from Theorem 3, evaluated at its selected outcomes and external parent settings. IDENTIFY returns an expression for this kernel from the supplied source. Substitution in Eq. (37) proves the returned formula.

Necessity: inconsistent normalized factors. Assign unqueried nodes fresh distinct labels, retaining all named query and intervention labels. If two natural nodes of the same primitive remain, their complete selected parent rows differ: otherwise every differing slot is fixed by the query or an intervention to the same value, and another allowed merge remains. The repeated-primitive construction in the proof of Lemma 1 gives two models equal on all of $\mathbf{P}_\star$ and different on E. In particular, this case is nonidentifiable from the supplied subset.

Otherwise K has at most one natural node of each primitive. Each entry in the consistency test is now either its unique natural-value symbol or a named constant. The test passes on the fresh assignment exactly when these entries agree for every assignment extending E_K. If it fails, consider the complete selected event $\{K = a\}$. It uses one controlled row and one output per primitive, so it is not a graph-forced zero. The inconsistency is a conflicting-settings c-component in the normalized counterfactual graph of its parent-completed, unnested conjunction. Shpitser and Pearl [23, Theorems 31(a)–32] provide two models equal on $\mathbf{P}_\star$ and different on this complete event. The

normalized-graph parent-record construction in the proof of Lemma 1 transfers the difference to the original event E. If positive models are required, first apply common private reset switches to this full-$\mathbf{P}_\star$ pair, and then the positive record extension. Full-$\mathbf{P}_\star$ equality survives the reset because each new hard row is a mixture of old rows with additional interventions. The nonzero contrast survives sufficiently small reset and record noise.

Necessity: an unavailable consistent kernel. Suppose every normalized factor is consistent and a component $\mathbf{C}$ is not identified by the source loop. Its underlying primitives are distinct. Write the supplied kernels as $Q[\mathbf{A}_j]$, with $\mathbf{A}_j = \mathbf{V} \setminus \mathbf{Z}_j$. By Kivva et al. [12, Theorem 1], failure from every source containing $\mathbf{C}$ gives positive observational models $\mathcal{M}_1, \mathcal{M}_2$ with equal complete supplied kernels and different $Q[\mathbf{C}]$. The witness can be localized. In the full version (arXiv:2206.01081v1), Section 5, Eqs. (6)–(9), only the law of one pairwise root U_0, with a child in $\mathbf{C}$, changes; all observed conditional mechanisms and every other root law agree. When no source contains $\mathbf{C}$, their Appendix C, Eqs. (27)–(29), changes only one observed mechanism in $\mathbf{C}$. Common finite private response tables implement the shared conditional mechanisms. Graph arguments omitted by the source construction can be restored identically.

Let $\mathbf{S}$ be the primitive names of K. Any second child of U_0 in $\mathbf{S}$ must also belong to $\mathbf{C}$, since otherwise its bidirected edge would join two distinct c-components of $\mathcal{G}[\mathbf{S}]$. In the alternative construction, the only changed observed mechanism lies in $\mathbf{C}$ and therefore does not occur in any companion factor. In either case every other parent-completed factor uses the same equations and root laws in both models. Paths through observed ancestors do not change this conclusion: its observed parent arguments have been fixed.

We can align the contrasting $Q[\mathbf{C}]$ cell with the named values required by the query. A consistent factor uses only one required value of each variable in $\mathbf{C} \cup Pa(\mathbf{C})$. Refine each alphabet by a finite surjection π_V to the source alphabet, sending its required label to the source contrast value and retaining all distinct named labels as distinct new states. Generate the original equation from projected parents, followed by a common positive private channel within the output's π_V-fiber. Every complete refined hard cell equals its projected old cell times the same positive product of channel weights. Hence the supplied kernels remain equal, the selected $Q[\mathbf{C}]$ contrast remains nonzero, and the localization is preserved.

Complete the assignment a on the remaining natural nodes. Every other factor is a consistent, unchanged controlled-kernel cell. It is positive: for a full assignment v, consistency makes the observational event $\{\mathbf{V} = v\}$ a subset of the corresponding hard-intervention event, whose probability is therefore at least $P(\mathbf{V} = v) > 0$. Thus

$$P_1(K = a) - P_2(K = a) = \big(Q_1[\mathbf{C}](a) - Q_2[\mathbf{C}](a)\big) \prod_{\mathbf{D} \neq \mathbf{C}} Q[\mathbf{D}](a) \neq 0. \tag{38}$$

Different factors may have different external settings; each factor's arguments in this display are those prescribed by $p_n(a)$. The parent-record construction now applies to this complete-cell contrast. It pads each original equation, not individual counterfactual copies. For every enlarged complete supplied row its probability is the old projected row times the common positive record-channel factors. This identity holds also for added intervention values outside the named label injections, so no unavailable experiment is assumed. At zero record noise the original-query contrast equals Eq. (38); it remains nonzero for sufficiently small positive noise. We have therefore constructed a positive pair with the same supplied input and different original-query probabilities. This proves the failure implication and completeness. □

The unconditional routine uses the IDENTIFY [27] algorithm (Algorithm 3). This algorithm is written in terms of *c-factors*, usually denoted as $Q[\mathbf{C}]$, for some set $\mathbf{C} \subseteq \mathbf{V}$. Such a factor is equivalent to $P_{\mathbf{V}\setminus\mathbf{C}}(\mathbf{C})$. When calling IDENTIFY with arguments $(\mathbf{C}, \mathbf{T}, Q[\mathbf{T}] = P_{\mathbf{V}\setminus\mathbf{T}}(\mathbf{T}), \mathcal{G})$, the algorithm determines if $P_{\mathbf{V}\setminus\mathbf{C}}(\mathbf{C})$ can be computed as a function of $P_{\mathbf{V}\setminus\mathbf{T}}(\mathbf{T})$ in $\mathcal{G}$.

D Conditional Counterfactuals

D.1 Conditional identification on finite state spaces

Fix an acyclic directed mixed graph G on the observed variables V. Each variable v has a specified finite state space $\mathcal{X}_v$. The structural models have independent exogenous roots with the incidences

Algorithm 3 IDENTIFY$(\mathbf{C}, \mathbf{T}, Q, \mathcal{G})$

Input: $\mathbf{C} \subseteq \mathbf{T} \subseteq \mathbf{V}$, $Q = Q[\mathbf{T}] = P_{\mathbf{V}\setminus\mathbf{T}}(\mathbf{T})$ and graph $\mathcal{G}$. Assuming $\mathcal{G}_{[\mathbf{C}]}$ and $\mathcal{G}_{[\mathbf{T}]}$ are composed of a single c-component.
Output: Expression for $Q[\mathbf{C}] = P_{\mathbf{V}\setminus\mathbf{C}}(\mathbf{C})$ in terms of Q or FAIL.

Let $\mathbf{A} \leftarrow An(\mathbf{C})_{\mathcal{G}_{[\mathbf{T}]}}$.
if $\mathbf{A} = \mathbf{C}$ **then return** $Q[\mathbf{C}] = \sum_{\mathbf{t}\setminus\mathbf{c}} Q$.
if $\mathbf{A} = \mathbf{T}$ **then return** FAIL.
if $\mathbf{C} \subsetneq \mathbf{A} \subsetneq \mathbf{T}$ **then**
    Let $\mathbf{T}'$ be the c-component containing $\mathbf{C}$ in $\mathcal{G}_{[\mathbf{A}]}$.
    Compute $Q[\mathbf{T}']$ from $Q[\mathbf{A}] = \sum_{\mathbf{t}\setminus\mathbf{a}} Q$.
    return IDENTIFY$(\mathbf{C}, \mathbf{T}', Q[\mathbf{T}'], \mathcal{G})$.
end if

permitted by G, including a private root at each observed variable. The constructions below use only private roots and roots on existing bidirected edges; no common root for an entire district is introduced. All models under consideration have a strictly positive observational distribution. Variables with singleton state spaces may first be replaced by their sole values throughout the equations, experiments, and query. Thus we assume $|\mathcal{X}_v| \geq 2$ in the coordinate constructions.

The target is a specified cell $P(A \mid B)$, where A and B are finite events formed from hard interventions and natural substitutions. Write $N = P(A \cap B)$ and $D = P(B)$, and restrict the target to $D > 0$. Numerator and evidence events are evaluated separately; an equality implied only by A is not imposed when evaluating B. For $S \subseteq V$, write

$$Q[S](x) = P_{x_{V\setminus S}}(x_S), \qquad Q[\varnothing] = 1.$$

A supplied complete experiment $Q[S]$ includes every setting of $x_{V\setminus S}$ and every output cell x_S. The collection of all such experiments is denoted by P_*.

D.2 Response tables and connected cumulants

Let $\mathcal{R}_v$ be the finite set of deterministic functions from $\mathcal{X}_{\mathrm{pa}_G(v)}$ to $\mathcal{X}_v$. The response table $R_v \in \mathcal{R}_v$ records the structural response of v at every observed-parent assignment. Every finite hard or naturally nested expression is a deterministic function of the tuple $R = (R_v : v \in V)$, evaluated by recursive substitution. Different occurrences of the same primitive use the same table. This representation therefore retains all equality constraints between repeated responses.

Let $G^{\leftrightarrow}$ be the graph containing only the bidirected edges. If $G^{\leftrightarrow}[S]$ is disconnected, the corresponding components of R_S depend on disjoint collections of independent roots. Consequently, their joint distribution factors over those components. This conclusion also holds for graph-compatible roots with more than two children, since their observed children form a bidirected clique.

Choose the uniform distribution p_v^0 on $\mathcal{R}_v$, and a rational basis $1, h_{v,1}, \ldots, h_{v,r_v}$ of functions on $\mathcal{R}_v$, where $r_v = |\mathcal{R}_v| - 1$ and $E_{p_v^0} h_{v,k} = 0$. Choose centered dual functions $h_{v,k}^*$ satisfying

$$E_{p_v^0}(h_{v,k} h_{v,l}^*) = \mathbf{1}\{k = l\}.$$

For each nonempty bidirected-connected set C and index tuple k_C, put

$$\theta_{C,k_C} = \mathrm{cum}\big(h_{v,k_v}(R_v) : v \in C\big).$$

Singleton cumulants are means, and cumulants indexed by disconnected sets are zero. The moment–cumulant identity becomes

$$E \prod_{v\in S} h_{v,k_v}(R_v) = \sum_{\substack{\pi \text{ a partition of } S \\ G^{\leftrightarrow}[C] \text{ connected for every } C\in\pi}} \prod_{C\in\pi} \theta_{C,k_C}. \tag{39}$$

Products of the one-table bases span all functions of R. Thus every response-table cell, and in particular N and D, is a polynomial in the connected cumulants θ. The baseline gives independent uniform outputs at all rows of each table. These polynomial identities hold for every compatible structural model, regardless of its original latent state spaces.

Lemma 6. *An open neighborhood of $\theta = 0$ is realized by finite structural models using only private roots and the bidirected edges of G. Every realizing model can be chosen to assign positive probability to every tuple of response tables.*

Proof. For every connected C with $|C| \geq 2$, choose a spanning tree T_C in $G^{\leftrightarrow}[C]$. For each (C, k_C) and every edge e of that tree, introduce an independent fair sign $U_{C,k_C,e}$. Signs on the same original bidirected edge may be bundled into one finite root with exactly that edge's two observed children. At $v \in C$, let

$$\sigma_{v,C,k_C} = \prod_{e \in T_C : e \ni v} U_{C,k_C,e};$$

for a singleton C, use $\sigma_{v,C,k_C} = 1$. Conditional on the incident signs, draw the table at each vertex privately with probabilities

$$p_v(r \mid U) = p_v^0(r) \left[1 + \sum_{C,k_C : v \in C} \alpha_{v,C,k_C} h^*_{v,k_v}(r) \sigma_{v,C,k_C} \right]. \tag{40}$$

Centering makes these probabilities sum to one. All brackets are strictly positive when the coefficients are sufficiently small, since there are only finitely many terms.

In the product expansion of (40), selecting one sign collection (C, k_C) at a proper nonempty subset of C gives expectation zero: an edge of its spanning tree crosses that subset, so its independent sign occurs exactly once. A surviving collection is therefore selected at every vertex of C. The additive bracket permits at most one selection at each vertex; surviving connected sets are consequently vertex-disjoint. Centering and duality now give (39), with

$$\theta_{C,k_C} = \prod_{v \in C} \alpha_{v,C,k_C}. \tag{41}$$

For each (C, k_C) choose a distinguished vertex v_C. Set all its other coefficients to a small positive ϵ, and set $\alpha_{v_C,C,k_C} = \theta_{C,k_C} / \epsilon^{|C|-1}$. For singleton sets, this simply sets $\alpha = \theta$. First choose ϵ so the absolute contribution of all fixed terms in every bracket is below $1/4$; then choose an open box around zero for θ so the variable terms contribute less than $1/4$. All conditional table probabilities remain strictly positive in that box.

The conditional table draws require only finite private roots. At each vertex, privately sample a vector containing one candidate table for every possible incident-sign configuration, using the corresponding probabilities in (40), and select the candidate at the realized configuration. The structural equation is $v = R_v(\mathrm{pa}_G(v))$. This implements the construction with independent finite roots. At zero the joint table distribution is the product of the p_v^0; throughout the box every joint table tuple has positive probability. □

D.3 The information supplied by all hard experiments

For a parent assignment a and output b, define the one-row indicator $g_{v,a,b}(R_v) = \mathbf{1}\{R_v(a) = b\}$. A selection of one such indicator per primitive in C is *globally consistent* if one assignment x to V supplies all its queried outputs and all its parent-row entries simultaneously. For every subset S of a globally consistent selection,

$$E \prod_{v \in S} g_{v,x_{\mathrm{pa}(v)},x_v}(R_v) = Q[S](x). \tag{42}$$

Indeed, under the indicated intervention the structural equations produce the complete cell x_S exactly when every selected row has its specified output; this follows in a topological order.

Expand each indicator as a constant plus a linear combination of the $h_{v,k}$. For $|C| \geq 2$, its joint cumulant is linear in θ_{C,k_C}, because cumulants containing a constant vanish. For a singleton, subtract its known baseline mean to obtain the same linear form. Let $L\theta$ collect these linear forms over all connected globally consistent selections. All entries of $L\theta$ are polynomials in P_* by cumulant inversion of (42). Conversely, every $Q[S](x)$ is reconstructed by the moment–cumulant formula from the same entries; disconnected cumulants vanish. We have therefore proved

$$P_*(M_1) = P_*(M_2) \quad \Longleftrightarrow \quad L\theta(M_1) = L\theta(M_2). \tag{43}$$

No independence between different rows of one table is assumed.

Choose a row basis L_0 of L, and complete $s = L_0\theta$ to invertible linear coordinates (s, w). The coordinates s are the full-experiment information; w consists of the remaining coordinates. Write the exact event polynomials in these coordinates as $N(s, w)$ and $D(s, w)$. If D is identically zero, the evidence is impossible. Otherwise cancel the polynomial greatest common divisor:

$$N = g\bar{N}, \qquad D = g\bar{D}, \qquad \gcd(\bar{N}, \bar{D}) = 1. \tag{44}$$

All these operations can be performed over the rational numbers.

Theorem 7. *On the specified finite state spaces, $P(A \mid B)$ is identifiable from P_* if and only if both $\bar{N}$ and $\bar{D}$ are independent of w. On success, $\bar{N}(s)/\bar{D}(s)$ is defined at every compatible model with $D > 0$. On failure, finite compatible models with strictly positive observational distributions and positive evidence agree on P_* but disagree on the specified conditional cell.*

Proof. If both reduced polynomials depend only on s, they give a function of P_* by (43). The identity $D = g\bar{D}$ implies $\bar{D} \neq 0$ at every actual point with $D > 0$, so the formula is valid there, including nongeneric points.

Over a field of characteristic zero, a reduced rational function depends on a coordinate w_j precisely when

$$(\partial_{w_j}\bar{N})\bar{D} - \bar{N}(\partial_{w_j}\bar{D})$$

is a nonzero polynomial. For completeness, if this expression vanishes, coprimality forces $\bar{N}$ to divide $\partial_{w_j}\bar{N}$ and $\bar{D}$ to divide $\partial_{w_j}\bar{D}$; comparing degrees gives zero derivatives. The constant-zero case is immediate. Thus if the criterion fails, such a nonzero polynomial exists.

An invertible linear coordinate change preserves the open set in Lemma 6. Whenever B is possible, at least one tuple of deterministic response tables satisfies it, so its probability is positive throughout that set. A nonzero polynomial cannot vanish on this open set. Choose a point where the displayed derivative numerator is nonzero and vary w_j slightly while holding s fixed. Both points have finite realizations from the lemma, with positive observational distributions and evidence. Equation (43) gives equal full experiments, whereas the conditional ratios differ. □

On success, substitute the polynomial expressions for s obtained from (42) into the reduced polynomials. This supplies polynomials $F(Q)$ and $J(Q)$ such that

$$\frac{N}{D} = \frac{F(Q)}{J(Q)}, \qquad J(Q) \neq 0 \quad \text{whenever } D > 0. \tag{45}$$

This statement concerns the ratio itself; its numerator and denominator need not be separately identifiable.

D.4 Coordinates for complete experimental kernels

Fix one reference state 0_v and put $\mathcal{X}_v^* = \mathcal{X}_v \setminus \{0_v\}$. For every nonempty bidirected-connected C, define

$$H_C = \{v \in C : v \text{ has no child in } G[C]\}, \qquad B_C = C \setminus H_C, \qquad E_C = \mathrm{pa}_G(C) \setminus C.$$

Thus H_C consists of the directed sinks in $G[C]$. For $h \in \mathcal{X}_{H_C}^*$ and arbitrary assignments b, e, define

$$q_C(h, b, e) = \frac{Q[C](h, b \mid e)}{Q[B_C](b \mid e)}. \tag{46}$$

The notation on the right suppresses intervention coordinates on which these kernels do not depend. Since H_C consists of sinks, summing it out of $Q[C]$ gives $Q[B_C]$. For an intrinsic set this is the published nested parameter: condition on its nonhead vertices and intervene only on its external parents [8, Theorems 5.4–5.5 and Appendix C].

Lemma 7. *The full array Q is a polynomial function $\Psi(q)$ of these connected corner tables. On positive full arrays,* (46) *is its rational inverse. The polynomial construction obeys parent dependence, sink marginalization, and factorization over bidirected components as formal identities.*

Proof. Proceed in increasing set size, starting from $Q[\varnothing] = 1$. For disconnected C, use $Q[C](x) = \prod_{K \in \mathrm{dis}(G[C])} Q[K](x)$. This follows by independence of the root collections entering the different components when all selected responses are evaluated at x. For connected C, first define cells whose head values are all nonreference by $Q[C](h, b \mid e) = q_C(h, b, e)Q[B_C](b \mid e)$. All remaining cells follow by inclusion–exclusion. Explicitly, let $Z = \{v \in H_C : x_v = 0_v\}$. Replace each reference indicator by one minus the sum of its nonreference indicators to obtain

$$Q[C](x) = \sum_{T \subseteq Z} (-1)^{|T|} \sum_{z_T \in \mathcal{X}_T^*} Q[C \setminus (Z \setminus T)](x_{C \setminus Z}, z_T).$$

The common external-parent assignment is suppressed. The term $T = Z$ uses the already defined nonreference-head cells; every other term uses a proper subset of C. These recursions are polynomial.

Sink marginalization holds by inclusion–exclusion in the connected case. In the disconnected case a sink is absent from the parent arguments of all other components, so it may be summed in its own component. Induction gives compatible repeated sink marginals and normalization. Parent dependence is preserved at every step, and component factorization is imposed directly. The recursion agrees with actual kernels by the same identities; dividing its nonreference-head cells by the positive sink marginal recovers every input corner coordinate. □

Let $\mathcal{A}$ be the supplied collection of free sets. For $S \in \mathcal{A}$, consider the conditional mixed graph with random vertices S, fixed vertices $V \setminus S$, and all incoming edges to the fixed vertices removed. Let $\mathcal{I}_G(S)$ be its intrinsic sets, namely districts obtainable in reachable conditional graphs by valid fixing operations. A random vertex is fixable when its district contains none of its proper directed descendants; fixing moves it to the fixed set and removes incoming directed edges and incident bidirected edges. Write $U = \bigcup_{S \in \mathcal{A}} \mathcal{I}_G(S)$, and split the full corner vector as $q = (u, z)$, where $u = (q_C : C \in U)$.

Lemma 8. *For positive compatible kernels, equality of all supplied $Q[S]$ is equivalent to equality of u. Each supplied kernel is polynomial in its intrinsic corner tables, and each such table is rationally recoverable from that kernel using positive denominators.*

Proof. Apply the discrete nested parameterization to each source conditional graph [8, Theorems 5.4–5.5 and Appendix C]. Its intrinsic parameter has head H_C and tail $B_C \cup E_C$, and its causal interpretation is exactly (46). Hence a set intrinsic in two sources gives the same controlled quantity in both; it does not introduce two independent parameters. Polynomial reconstruction gives one implication, and positive rational extraction gives the other. The extraction divides only by positive kernel marginals and conditional probabilities. In particular, it does not replace conditioning on B_C by intervening on B_C. □

We also need actual models throughout an open set of these coordinates, with positive counterfactual evidence, not merely positive hard kernels. Lemma 11 supplies the following precise interface: near the uniform full hard array, it gives a continuous section $q \mapsto \sigma(\Psi(q))$ into the connected response-table cumulants, preserves the whole array $\Psi(q)$, and takes the uniform corner vector to the product response-table baseline. Combining it with Lemma 6, every corner vector in a sufficiently small open neighborhood has a finite compatible realization assigning positive mass to every response-table tuple. Thus every possible original evidence event is positive there. This use of the section is independent of any conditional identification claim: the section is constructed from the full-kernel identities in Lemma 7 and the linear information map.

D.5 Identification from the supplied experiments

First apply Theorem 7. Its negative pair also agrees on any supplied subset of experiments. On success, take F, J from (45), substitute the corner reconstruction, and reduce once more:

$$F(\Psi(q)) = h(q)a(q), \qquad J(\Psi(q)) = h(q)b(q), \qquad \gcd(a, b) = 1. \tag{47}$$

Theorem 8. *On the specified finite state spaces, the original conditional cell is identifiable from the supplied complete experiments if and only if the full-experiment test succeeds and both a and b are independent of the unused corner coordinates z. On success the answer is $a(u)/b(u)$, with u recovered from the sources. It is defined at every positive observational model with $P(B) > 0$. On*

failure there are finite compatible models with positive observation and positive original evidence that agree on every supplied kernel cell and disagree on the requested conditional cell.

Proof. Suppose first that a, b depend only on u. For every observed assignment x, consistency gives $P_{x_{V\setminus S}}(x_S) \geq P(x) > 0$. Thus all full kernels and all required corner denominators are positive. Lemma 8 recovers u from the inputs. Equations (45) and (47) give the answer. Moreover, $D > 0$ implies $J(Q) \neq 0$ by the first equation, and the second equation then implies $b(q) \neq 0$. The two exact polynomial factorizations therefore protect the output denominator at every actual point of the target's domain.

Conversely, assume the full-experiment test succeeds but the reduced ratio depends on some z_j. As in Theorem 7, $(\partial_{z_j} a)b - a(\partial_{z_j} b)$ is a nonzero polynomial. It is nonzero somewhere in the open corner neighborhood supplied above. Every point in that neighborhood has an actual finite response-table realization with positive original evidence, and $J(\Psi(q)) \neq 0$ there. Vary z_j slightly while holding all u fixed. The two realizations agree on the complete supplied inputs by Lemma 8, while their ratios differ. The full-experiment failure case was already handled by its stronger counterpair. □

Impossible evidence is reported as undefined before either ratio test. The cases $N = 0$ and $N = D$ give the constants zero and one. The two theorems are finite exact algebraic procedures on specified alphabets; they make no claim of polynomial running time or of a compact graphical expression. A graph-uniform decision additionally requires the finite alphabet arguments given separately.

D.6 Finite alphabets and conditional identification

We consider one specified conditional cell

$$R = P(A \mid B) = \frac{N}{D}, \qquad N = P(A \cap B), \quad D = P(B),$$

on its domain $D > 0$. The events may contain finite hard or natural nested counterfactuals. All named values and their equality relations are retained. The observed state spaces are finite, but their cardinalities are not fixed in the graph-uniform identification question. Write $\mathcal{V}$ for the observed vertex set. The supplied inputs are complete hard-intervention kernels $Q[S]$, including all settings of their intervention variables.

We use the following fixed-alphabet results established above. First, let R_V denote the complete response table of primitive V. Connected cumulants of functions of the R_V have an open set of actual finite-root realizations around a strictly positive product-table distribution. The roots have only graph-permitted incidences. Equality of all hard kernels is equivalent to equality of a linear map $L\theta$ of these response cumulants. For the exact numerator and evidence polynomials, cancellation of their polynomial greatest common divisor gives a complete full-experiment test: the reduced polynomials must depend only on $L\theta$. Its successful denominator is nonzero at every actual model with $D > 0$.

Second, for every bidirected-connected nonempty set C, put

$$H_C = \{v \in C : v \text{ has no child in } G[C]\}, \qquad B_C = C \setminus H_C, \qquad E_C = \mathrm{Pa}_G(C) \setminus C.$$

Choose a reference value at each primitive. The connected corner tables are

$$q_C(h, b, e) = \frac{Q[C](h, b \mid e)}{Q[B_C](b \mid e)}, \qquad h_v \neq 0_v \quad (v \in H_C).$$

There is a polynomial reconstruction $Q = \Psi(q)$ of the full hard array. For each supplied source S, its positive kernel is equivalent to the corner blocks indexed by the intrinsic sets of its source graph. These blocks are rationally recoverable using positive denominators. The same block has the same meaning when recovered from different sources. These statements use conditioning on B_C and interventions on E_C; they do not replace the conditioning by interventions on the whole tail.

D.7 A bound for the full-experiment problem

Compile the numerator and evidence separately into finite acyclic expression graphs whose gates are evaluations of original structural response functions. Include every recursive natural-ancestor

evaluation. Let their gate counts be m_N, m_D, and put $m = \max(m_N, m_D)$. Counting repeated gates separately is allowed. Let c_V count the distinct values of V named anywhere in the query.

For a fixed observed alphabet, choose a reference 0_V. For a parent row a and $b \neq 0_V$, write

$$e_{V,a,b} = \mathbf{1}\{R_V(a) = b\}, \qquad H_{V,T,b} = \prod_{a \in T} e_{V,a,b(a)},$$

where T is a nonempty set of distinct parent rows. The functions 1 and $H_{V,T,b}$ form a tensor-product basis for functions on the response-table space of V. For a connected primitive set C, use the coordinates

$$\theta_{C,(H_V)} = \mathrm{cum}(H_V(R_V) : V \in C), \qquad w(\theta_{C,(H_V)}) = \sum_{V \in C} |T_V|.$$

Singleton cumulants are means. This basis need not be centered: its change from a centered basis is invertible affine on singleton coordinates and invertible linear on each higher-order connected block. It therefore retains the open realization property.

Lemma 9 (Weighted degree and hidden directions)**.** *The numerator and evidence polynomials have weighted degrees at most m_N and m_D. Moreover,* $\ker L$ *has a basis consisting of coordinate vectors $\mathbf{e}_\alpha$ and differences $\mathbf{e}_\alpha - \mathbf{e}_\beta$. In the latter case, the two coordinates use the same connected primitive set and the same single parent row at every primitive, and have equal weight.*

Proof. Expand an event indicator over assignments to its response-expression gates. Each satisfying trace contributes at most one response-row indicator per gate, and each deterministic table tuple produces exactly one trace. Replace a reference-output indicator by $1 - \sum_{b \neq 0_V} e_{V,a,b}$. Equal outputs at an identical row combine by idempotence; unequal outputs give zero. Neither operation increases the number of row tests. Regroup by primitive and apply the moment–cumulant partition formula. Each resulting product of connected cumulants has the same total weight as its basis moment, proving the degree bound.

A visible source cumulant uses at most one parent row per primitive. Coordinates containing a multirow basis function are individually hidden. For the remaining coordinates, L separates by connected set C and selected parent-row tuple $(a_V)_{V \in C}$. Singleton one-row blocks are fully observed. In a larger block, incompatible prescriptions for a common parent make the entire block hidden. Otherwise, every vertex V having a child inside C has a forced output t_V, determined by those child rows; sink outputs are unrestricted. In the nonreference output basis, the forced output has coefficients

$$\lambda_V(b) = \begin{cases} \mathbf{1}\{b = t_V\}, & t_V \neq 0_V, \\ -1, & t_V = 0_V, \end{cases} \qquad b \neq 0_V.$$

Fix a nonreference output tuple for the sinks. The visible row space on the remaining output tuples is spanned by the single vector $\prod_V \lambda_V(b_V)$ over the nonsinks. Its entries are zero or one common nonzero sign. Its kernel is spanned by zero-column coordinate vectors and differences between nonzero-column coordinate vectors. This proves the assertion, including blocks with no nonsinks. □

Lemma 10 (Restriction of observed state spaces)**.** *Let $\mathcal{B}_V \subseteq \mathcal{X}_V$ contain 0_V and every named value. The numerator and evidence polynomials on $\mathcal{B}$ are obtained from those on $\mathcal{X}$ by retaining identically indexed cumulant coordinates and setting all other coordinates to zero. A hidden direction supported on retained coordinates remains hidden. Conversely, a small-domain hidden direction extended by zero remains hidden on the larger domains.*

Proof. Embed a small response table into the large table space by retaining its values on parent rows inside $\mathcal{B}$ and returning 0_V on every other row. Each basis function $H_{V,T,b}$ becomes its identical small-domain function or zero. The same is true of its connected cumulant coordinate. No distinct surviving coordinates are identified. All named values are retained, so execution of the query is unchanged. The asserted polynomial identities follow from the basis expansion, or from equality on the small-domain open realization set.

A visible cumulant restricts to a small-domain visible cumulant or zero. An outside row becomes a constant, which annihilates a higher-order cumulant; an outside output on an inside row gives zero.

If every row and output survives, the globally consistent assignment extends inside $\mathcal{B}$ by assigning reference values to unrelated primitives. Singleton constants are known and have zero linear part. Every small-domain visible row is also the restriction of its large-domain counterpart. These two facts prove both hidden-direction assertions. The deterministic embedding is used only for polynomial identities; positive countermodels are obtained from the open realization set. □

Theorem 9 (Finite observed-alphabet bound)**.** *If the specified conditional cell is nonidentified from all hard experiments on some finite alphabets, it is nonidentified on alphabets satisfying*

$$|\mathcal{X}_V| \le c_V + 3m + 1 \qquad \textit{for every } V.$$

It suffices to perform the full-experiment test once at exactly these sizes.

Proof. By the fixed-alphabet criterion, there is $h \in \ker L$ such that

$$\Delta_h = (\partial_h N)D - N(\partial_h D)$$

is a nonzero polynomial. Its linearity in h permits a sparse basis direction from Lemma 9. Let ℓ be the common weight of its one or two supported coordinates. At least one occurs in N or D, so $\ell \le m$. Every monomial of Δ_h has weight at most $2m - \ell$. Choose a monomial with nonzero coefficient and retain all row tests indexing its coordinates, together with the tests indexing both coordinates of h. There are at most $(2m - \ell) + 2\ell \le 3m$ such tests, with multiplicity. Each test contains at most one value of a given primitive: its output, or its value in the parent row. Acyclicity excludes a self-parent. Adding the named values and a reference gives the claimed bound.

The selected monomial and the direction h survive domain restriction. Differentiation along h commutes with that restriction. Every other monomial survives unchanged or vanishes; none becomes the selected monomial. Its nonzero coefficient therefore survives. The smaller-domain derivative is nonzero in a hidden direction. The open realization theorem supplies two actual finite-root models with equal full hard kernels, positive observation and evidence, and different values of the originally specified conditional cell.

For completeness, evidence positivity causes no obstruction. A nonzero derivative implies a nonzero evidence polynomial. Its event has positive probability at a strictly positive product-table baseline, and the nonzero derivative cannot vanish on an open neighborhood. To enlarge a witness to exactly the stated sizes, extend its hidden direction by zero. Lemma 10 makes it hidden on the enlarged domains, and its derivative restricts to the old nonzero polynomial. Apply the open realization argument again. □

An impossible-evidence verdict at the bounded sizes is also graph-uniform. Any satisfying execution on any finite alphabet uses at most m_D returned values, besides named constants. Retaining these values and reproducing their response rows gives a satisfying execution below the same bound. Queries with no response gates are decided directly as constant events.

D.8 A uniform section of complete experimental information

For now assume that every observed domain has at least two states. Choose distinct $0_V, 1_V$, using named values where possible. For $|C| \ge 2$, define

$$\kappa_C(x) = \sum_{\pi \text{ partition of } C} (|\pi| - 1)!(-1)^{|\pi|-1} \prod_{J \in \pi} Q[J](x_J \mid x_{\mathcal{V}\setminus J}).$$

This is the ordinary cumulant of the globally consistent single-row response indicators. For singletons use their ordinary means $Q[\{V\}]$. The higher-order arrays vanish on disconnected sets, depend only on $C \cup \mathrm{Pa}(C)$, and sum to zero in each sink of $G[C]$. The moment–cumulant inverse reconstructs the full hard array.

Define a section σ assigning response cumulants to a full hard array. A singleton one-row mean is assigned its given $Q[\{V\}]$ value. A singleton basis function involving $k \ge 2$ distinct parent rows is assigned the independent uniform-row moment d_V^{-k}, where $d_V = |\mathcal{X}_V|$. A cross-primitive cumulant containing any multirow basis function is zero. For the remaining case, select one parent row per primitive of connected C. Incompatible repeated assignments within the parent-row addresses give zero. This test does not reject an output y_I merely because it differs from b_I: those off-diagonal

contrast terms are required. Otherwise put $I = C \setminus H_C$, let b_I be the outputs forced by the selected child rows, and let e be their consistent external settings. Set

$$z_V(y; b) = \mathbf{1}\{y = b\} - \mathbf{1}\{y = \mathrm{alt}_V(b)\}, \qquad \mathrm{alt}_V(b) = \begin{cases} 1_V, & b = 0_V, \\ 0_V, & b \neq 0_V. \end{cases}$$

The complete one-row cumulant array is assigned

$$\theta_C(y_C; \text{selected rows}) = \kappa_C(b_I, y_{H_C}; e) \prod_{V \in I} z_V(y_V; b_V).$$

Its nonreference output cells specify the basis coordinates.

Lemma 11 (Visible-cumulant section)**.** *The section preserves every complete hard kernel. At the uniform hard array it equals the strictly positive product distribution of independent uniform response-table rows. Every sufficiently nearby $\sigma(\Psi(q))$ has an actual finite-root realization positive on every response-table tuple.*

Proof. Each z_V sums to zero and equals one at its forced value. The assigned array therefore sums to zero in every output coordinate; for sinks this uses the sink restriction on κ_C. Its value at every globally consistent tuple equals the required κ_C, so all visible cumulants and all hard kernels are preserved. At the uniform array, cross-primitive cumulants vanish and all singleton moments agree with independent uniform rows. For fixed finite alphabets the section is continuous. Its composition with the polynomial corner reconstruction consequently lies in the open response-table realization set near the uniform corner. Every possible evidence event is positive throughout this neighborhood. □

Lemma 12 (Bounded query expressions)**.** *The expressions $a(Q) = N(\sigma(Q))$ and $b(Q) = D(\sigma(Q))$ have finite typed summation syntax depending only on the graph and query. After multiplication by $K(d) = \prod_V d_V^m$, their coefficients contain no cardinality denominators.*

Proof. Use the finite gate-trace expansion from Lemma 9. Partition its parent-row addresses by equality, discard incompatible outputs at identical rows, expand reference-output indicators, and apply the moment–cumulant formula after grouping tests by primitive. Only the rows requested by the trace are used; subsets of the entire response-table row space are never enumerated. The explicit section uses equality guards, fixed graph subsets and partitions, and coefficients d_V^{-k} with total exponent at most m for each primitive. This proves the denominator assertion. The polynomial corner reconstruction has bounded graph-based recursion. Substituting it adds only bounded products, typed sums and equality guards. Write each remaining d_V as a typed sum of 1 to obtain finite tensor expressions with rational constant coefficients. □

D.9 Arbitrary supplied experiments and a finite identity test

Let u collect the corner blocks recoverable from the supplied sources, and write the remaining blocks as ξ. Assume that the full-experiment graph-uniform test has succeeded and that the evidence is possible. Put $\widetilde{a} = Ka$ and $\widetilde{b} = Kb$. Using an independent copy ξ' of the unused corner blocks, form

$$F(u, \xi, \xi') = \widetilde{a}(\Psi(u, \xi))\widetilde{b}(\Psi(u, \xi')) - \widetilde{a}(\Psi(u, \xi'))\widetilde{b}(\Psi(u, \xi)).$$

For fixed alphabets, this polynomial vanishes identically exactly when the requested ratio depends only on the supplied corner blocks. Indeed, the section realizes an open neighborhood of actual models with positive evidence. Its ratio agrees there with the full-experiment identifying function, establishing their rational identity. If F is nonzero, two nearby corner vectors sharing u give actual positive-evidence countermodels with the same supplied kernels. If $F = 0$, the rational identifying function belongs to the field in u.

Lemma 13 (Finite tensor identity test)**.** *Expand F into closed typed summation templates. Let L_V bound the number of bound indices of type V in any template, including all expansions and both independent copies. Let t_V count its distinct named and distinguished labels. The identity holds on every nonsingleton finite alphabet if and only if it holds at every point of the finite product grid*

$$d_V \in \{d_V^0, \ldots, d_V^0 + L_V\}, \qquad d_V^0 = \max(2, t_V + L_V).$$

Proof. Fix a free-corner-table monomial using k_V nondistinguished labels of type V. Every such label comes from a bound index, so $k_V \leq L_V$. Partition assignments of the indices by their equality pattern with each other, the distinguished labels, and these k_V labels. An accepted pattern contributes a falling factorial in $d_V - t_V - k_V$, of degree at most $L_V - k_V$. There are finitely many patterns and monomial equality types, independently of the alphabet sizes. Every type occurs throughout the stated grid. If the polynomial is zero there, each coefficient polynomial vanishes at $L_V + 1$ sizes in every variable and hence vanishes identically. The same counting identity applies to smaller domains supporting the distinguished labels, where impossible patterns contribute zero. The converse is immediate. □

The count L_V is computed for this entire cross-product expression. It includes query sums, reference-output expansions, corner reconstruction, and the sums expressing $K(d)$; a count for the unconditional query alone does not suffice. A grid failure produces actual countermodels by Lemma 11, not merely signed kernel assignments.

D.10 Singleton domains and defined formulas

Lemma 14 (Singleton-domain reduction)**.** *For each admissible set S of singleton primitives, substitute their sole values recursively throughout the query and remove their mechanisms and graph vertices. Replace every supplied free set A by $A \setminus S$. This reduction preserves the numerator and evidence individually, as well as equality of the supplied kernels and observational positivity.*

Proof. Every singleton mechanism returns its sole value under every intervention. In the natural-response syntax remove intervention entries whose target lies in S, while retaining constant-valued entries targeting surviving primitives. Recursive substitution therefore preserves every surviving structural evaluation and both event indicators. Restricting roots and equations gives the reduced SCM; conversely, extend any reduced SCM by constant mechanisms at the deleted primitives. Both operations respect graph-permitted root incidences. Deleting the constant coordinates from each hard kernel and observational cell gives the asserted exact correspondences. □

Only primitives with at most one distinct named value can be singletons. Enumerate all admissible patterns S, at most $2^{|\mathcal{V}|}$ of them, and apply the nonsingleton construction to each residual problem. There choose $0_V, 1_V$ among named values where possible, so $t_V = \max(c_V, 2)$ before additional compiler labels are counted. Patterns with impossible evidence are skipped. If every pattern has impossible evidence, report the query as undefined. If no primitive remains, evaluate the constant events directly; a possible constant pattern needs no coordinate or grid construction.

The raw section ratio is used for decision only. For formula evaluation at the actual finite alphabets, the successful full-experiment test gives coprime polynomials $\overline{N}(s)/\overline{D}(s)$ in visible response cumulants. The identity $D = g\overline{D}$ makes $\overline{D}(s)$ nonzero at every actual input with positive evidence. Express s polynomially in the full hard kernels, substitute $Q = \Psi(q)$, and cancel the polynomial greatest common divisor again. The identity $F = 0$ makes both resulting coprime polynomials depend only on u. Their denominator is nonzero at every relevant actual input, because the denominator before this cancellation was already nonzero. Finally recover u from the positive supplied kernels using their positive extraction denominators. This also covers inputs where the raw section numerator and denominator both vanish.

Theorem 10 (Graph-uniform conditional identification)**.** *Under the stated fixed-alphabet response and corner results, the following finite procedure is sound and complete for the specified conditional cell from an arbitrary collection of complete hard experiments:*

1. *Enumerate the admissible singleton patterns and reduce them exactly; recompute the expression graphs, named values and gate counts after substitution.*
2. *Perform the bounded full-experiment test, handling impossible evidence and constant ratios separately. A failure is also a failure for the supplied subset.*
3. *For each remaining residual problem, construct the explicit section and cross-product F, and test its finite alphabet grid.*

4. *If every test succeeds, return the denominator-safe formula at the actual input alphabets and their actual singleton pattern.*

Every rejection has graph-compatible finite positive countermodels for the original named conditional cell. Every returned formula is defined throughout the cell's positive-evidence domain.

Proof. The full-experiment rejection and its alphabet bound follow from Theorem 9. Conditional on that test succeeding, the section, source-coordinate equivalence and Lemma 13 prove necessity and sufficiency for each nonsingleton residual problem. Lemma 14 transfers both decisions and countermodels to the original query, preserving N and D separately. The denominator argument above gives the asserted evaluation domain. □

This is an algebraic completion of the identification problem. It makes no polynomial running-time claim and does not assert that the original simpler conditional factor procedure has the same failure set. The universal decision uses finitely many bounded alphabets; its formula construction remains finite for each actual input alphabet.

Lemma 3 (Conditional Query Reduction)**.** *On fixed finite observed alphabets, write $N = g\bar{N}$ and $D = g\bar{D}$ in the polynomial coordinates (s, h), where g is a greatest common divisor and $\bar{N}, \bar{D}$ are coprime. The specified cell is identified from P^* if and only if both reduced polynomials are independent of h. On success their ratio is defined at every compatible input for which the original evidence has positive probability.*

For arbitrary supplied complete experiments, express that ratio in full intrinsic corner coordinates q, and again cancel the polynomial greatest common divisor. The cell is identified if and only if both resulting polynomials depend only on the corner blocks recoverable from at least one supplied experiment.

Proof. Apply the response-cumulant reduction and the intrinsic-coordinate reduction proved above. The two greatest-common-divisor identities protect the denominator on every positive-evidence input. The open response-table realization supplies actual same-input countermodels for every nonzero hidden derivative. □

Theorem 6 (CTFID completeness)**.** *In the positive, domain-flexible graph-uniform class, a specified conditional cell with finite hard/natural response syntax is identifiable from $\mathbb{Z}$ and $\mathcal{G}$ if and only if the revised* CTFID *returns an expression for it. The statement concerns its positive-evidence domain; universally impossible evidence is reported separately. Every negative decision has two finite graph-compatible models agreeing on all supplied complete kernels and disagreeing on the specified cell, with positive observational laws and positive evidence in both.*

Proof. The optional separate marginal calls are sound on the original positive-evidence domain. The bounded full-experiment test is necessary for every supplied subset. After it succeeds, the source comparison on every singleton pattern is sound and complete by Theorem 10. The final reduced formula is defined at every admissible input by the denominator argument in that theorem. Evidence impossible on the bounded full-experiment alphabet is impossible on every finite alphabet by the execution-trace argument above. □

E Corrected conditional running example

We retain the diagram in Fig. 7 and the original query

$$P(Y_{x_1 Z_{x_0}} = y_1 \mid X_{w_0} = x_1, T_{r_1} = t_1, R = r_0).$$

Since the only observed parent of Y is Z, setting $Z = z$ makes the additional intervention on X irrelevant to Y. After unnesting and using the evidence $R = r_0$, define

$$\begin{aligned} K_{t,z} &= P(Y_z = y_1, X_{w_0,t} = x_1, R = r_0), \\ J_t &= P(X_{w_0,t} = x_1, R = r_0), \\ L_z &= P(Z_{x_0,r_0} = z), \\ H_t &= P(T_{r_0} = t, T_{r_1} = t_1). \end{aligned}$$

The exact numerator and denominator of the original conditional are

$$N = \sum_{t,z} K_{t,z} H_t L_z, \qquad D = \sum_t J_t H_t. \tag{48}$$

The private noise of T is independent of the roots entering the other factors, but it is shared by T_{r_0} and T_{r_1}. After the displayed parent assignments, the roots entering Z are also disjoint from those entering the X, R, Y factor. These facts justify the factorization in Eq. (48).

The available experiments identify

$$\begin{aligned} K_{t,z} &= P_{w_0,t}(r_0) P_{w_0,t}(x_1 \mid r_0) P_{w_0,t}(y_1 \mid r_0, x_1, z), \\ J_t &= P_{w_0,t}(r_0) P_{w_0,t}(x_1 \mid r_0), \\ L_z &= P_{w',t'}(z \mid r_0, x_0), \end{aligned}$$

for arbitrary fixed w', t'. They do not generally identify the joint response factor H_t. Replacing it by the marginal $P(T_{r_0} = t)$ removes part of the evidence and changes the query.

A positive counterexample for $r_0 \neq r_1$. Let U_R, U_W, U_Z, B, U be independent fair bits, and let $E \sim \mathrm{Bernoulli}(1/4)$ be independent of them. In both models use

$$R = U_R, \quad W = U_W, \quad Z = U_Z, \quad X = T \oplus U \oplus E, \quad Y = U.$$

The two models differ only in the equation for T:

$$M_1 : \; T = B, \qquad M_2 : \; T = B \oplus R.$$

Here U is the root on the permitted $X \leftrightarrow Y$ edge; the other displayed roots are private. All other permitted arguments may be inactive. Thus both models are compatible with the diagram. For every observational cell,

$$P(r, w, t, z, y, x) = \frac{1}{32} P(E = x \oplus t \oplus y) > 0.$$

The models agree on every complete hard-intervention kernel. If T is intervened on, its differing equation is removed. Otherwise, for every natural or fixed value of R, its selected response is a fair bit independent of the remaining roots in both models. Every other structural equation and root distribution is the same. This proves equality for arbitrary intervention targets and values, including interventions on R, X, or Y.

Take

$$r_0 = 0, \quad r_1 = 1, \quad t_1 = x_1 = y_1 = w_0 = 0, \quad x_0 = 1.$$

The nested outcome equals U. On $R = 0$, both models have $T_{r_0} = B$, whereas $T_{r_1} = B$ in M_1 and $T_{r_1} = 1 - B$ in M_2. Therefore the evidence $T_{r_1} = 0$ sets $B = 0$ in M_1 and $B = 1$ in M_2. The additional evidence $X_{w_0} = 0$ implies $U = E$ in M_1 and $U = 1 - E$ in M_2. Consequently,

$$D_1 = D_2 = \frac{1}{8}, \qquad N_1 = \frac{3}{32}, \quad N_2 = \frac{1}{32}, \qquad \frac{N_1}{D_1} = \frac{3}{4}, \quad \frac{N_2}{D_2} = \frac{1}{4}.$$

The requested conditional cell is therefore nonidentified even with all hard experiments. The previously displayed unnormalized expression equals $1/8$ in both models; it omits the T_{r_1} evidence.

An identified comparison using $r_1 = r_0$. For the modified evidence $T_{r_0} = t_1$,

$$H_t = \mathbf{1}\{t = t_1\} P(T_{r_0} = t_1).$$

The common T factor cancels between the numerator and denominator in Eq. (48). Substituting the identified K, J, L expressions gives

$$\begin{aligned} &P(Y_{x_1 Z_{x_0}} = y_1 \mid X_{w_0} = x_1, T_{r_0} = t_1, R = r_0) \\ &\quad = \sum_z P_{w_0,t_1}(y_1 \mid r_0, x_1, z) P_{w',t'}(z \mid r_0, x_0). \end{aligned}$$

This is an identified query on the same diagram and from the same complete experiments. It is a different query from the original case $r_0 \neq r_1$.

Algorithm 4 GID*$(\mathbf{Y}_*, \mathbf{y}_*, \mathbb{Z}, \mathcal{G})$

Input: $\mathcal{G}$ a causal diagram, $(\mathbf{Y}_* = \mathbf{y}_*)$ a finite unnested counterfactual event; and available distribution specification $\mathbb{Z}$.
Output: $P(\mathbf{Y}_* = \mathbf{y}_*)$ in terms of available distributions specified by $\mathbb{Z}$ or FAIL if not identifiable from $\langle \mathcal{G}, \mathbb{Z} \rangle$.

if $\mathbf{Y}_* = \emptyset$ **then return** 1
if there exists $x \in \mathbf{y}_*$ for some $X_{x'}$ with $x \neq x'$ **then return** 0
if there exists $x \in \mathbf{y}_*$ for some X_x **then return** GID*$(\mathbf{Y}_* \setminus \{X_x\}, \mathbf{y}_* \setminus \{x\}, \mathbb{Z}, \mathcal{G})$
$(\mathcal{G}', (\mathbf{Y}'_*, \mathbf{y}'_*)) \leftarrow$ **make-cg**$(\mathcal{G}, (\mathbf{Y}_*, \mathbf{y}_*))$
if $\mathbf{Y}'_*$ is **INCONSISTENT then return** 0
Give each unqueried natural node a distinct fresh value symbol; let a denote this assignment together with the queried values. For each natural node n, let $p_n(a)$ be its inherited original parent row.
if $\mathcal{G}'$ has more than one c-component $\mathbf{S}_1, \ldots, \mathbf{S}_k$ **then**
  for each $\mathbf{S}_i$ **do**
    Let $\Gamma_i(a) = \bigwedge_{n \in \mathbf{S}_i} \{V_{n[p_n(a)]} = a_n\}$.
    Let $F_i(a) \leftarrow$ GID*$(\Gamma_i(a); \mathbb{Z}, \mathcal{G})$.
    if F_i is FAIL **then return** FAIL
  end for
  return $\sum_{a \supseteq (\mathbf{Y}'_* = \mathbf{y}'_*)} \prod_i F_i(a)$.
end if
if $\mathcal{G}'$ has a single c-component $\mathbf{S}$ **then**
  Let $\Gamma(a) = \bigwedge_{n \in \mathbf{S}} \{V_{n[p_n(a)]} = a_n\}$ and $\mathbf{C} = \{V_n : n \in \mathbf{S}\}$.
  if $\Gamma(a)$ is symbolically inconsistent **then return** FAIL
  Let $\mathbf{x}$ assign $\mathbf{V} \setminus \mathbf{C}$, agreeing with the external parent settings of $\Gamma(a)$; choose irrelevant settings arbitrarily.
  Let $F(a) \leftarrow$ GID$(a_{\mathbf{C}}, \mathbf{x}, \mathbb{Z}, \mathcal{G})$.
  if F is FAIL **then return** FAIL
  return $\sum_{a \supseteq (\mathbf{Y}'_* = \mathbf{y}'_*)} F(a)$.
end if

F Relation with Previous Algorithms

In Algorithm 4, we express the component decomposition of ID* [22] using the parent-completed factors of this paper. We retain value-aware equality normalization and use the corrected GID [12] to identify each consistent controlled kernel from the available input distributions $\mathbb{Z}$.

The event notation GID*$(\Gamma; \mathbb{Z}, \mathcal{G})$ abbreviates the original first two arguments by their conjunction. Here V_n denotes the primitive variable of node n, and $p_n(a)$ assigns each of its original parents separately. Thus $\Gamma_i(a)$ never clamps a primitive to two values in one world: it is a conjunction of ordinary, individually specified controlled responses.

The recursion retains the original diagram $\mathcal{G}$ and full input $\mathbb{Z}$. Each call is made symbolically, using distinct fresh labels for unqueried nodes, as in CTFIDU; the displayed sums are formed only after the symbolic factor formulas have been obtained. A successful formula is then evaluated for all assignments in its sum. The test is not a loop that rejects a numerical summand because one factor failed while another was zero. After completed value-aware normalization, the generic selected parent rows cannot coincide with incompatible outputs: such copies would already have merged and returned zero. Therefore the generic component calls do not return zero on a nonzero normalized query.

A recursive $\Gamma_i(a)$ queries every natural node in its component, so it reaches the single-component branch with no unqueried natural outputs. Its last sum is then vacuous. At the top level that sum is needed when a single c-component contains unqueried ancestors. Internal parent settings are removed by the controlled-kernel reduction of Theorem 3; the GID call targets $\mathbf{C}$ and intervenes only on its complement. The normalized graph is an event-preserving quotient, not a literal equality with the raw ancestor set.

Lemma 15. *After value-aware normalization, a component subproblem of* GID* *has the same parent-completed factor as the corresponding normalized component used by* CTFIDU. *A nonzero consistent component reduces to the same controlled kernel in both procedures.*

Proof. For a complete assignment a, topological substitution gives $\mathbf{1}\{\mathcal{G}' = a\} = \prod_n \mathbf{1}\{V_{n[p_n(a)]} = a_n\}$. Grouping these original equation events by c-component gives precisely $\Gamma_i(a)$, with the inherited parent slots; independent primitive root sets give the same factorization as Eq. (37). This construction is valid even when distinct nodes have the same primitive name, because each conjunct has its own ordinary parent intervention.

On the generic assignment, different remaining copies of one primitive have different parent rows after completed normalization, so they fail the same consistency test as in CTFIDU. If no repeated primitive remains, consistency makes internal parent settings agree with their component outcomes and all external settings agree for each external primitive. Theorem 3 then gives $Q[\mathbf{C}]$, evaluated at those settings. The source loop of CTFIDU and the corrected GID call identify this same kernel by Kivva et al. [12, Theorem 1]. Only after obtaining these formulas do the procedures sum over the original unqueried natural outputs. □

Corollary 2. *In the positive, domain-flexible graph-uniform class, an unnested counterfactual probability is identifiable from $\mathbb{Z}$ and $\mathcal{G}$ if and only if the corrected* GID* *returns an expression for it.*

Proof. Component expansion and equality reduction are sound by Shpitser and Pearl [23]; consistent source calls are sound by Kivva et al. [12]. For necessity, the normalized factors agree by Lemma 15. A repeated primitive or conflicting settings give the full-$\mathbf{P}_\star$ countermodels used in the proof of Theorem 5. Failure to identify a consistent component uses the same localized positive source countermodel and parent-record construction. Those are differences of the original query, so summation of the component expression does not invalidate the negative conclusion. □

Changes

Unofficial AI-corrected version, 2026-10-21. Author review pending. Prepared with AI assistance in a separate copy of the authors' June 2026 source. This is not a publisher-issued correction. Original and earlier working versions are preserved.

Original: Juan D. Correa, Sanghack Lee, and Elias Bareinboim, "Nested Counterfactual Identification from Arbitrary Surrogate Experiments," *Advances in Neural Information Processing Systems* 34 (2021), 6856–6867. Original proceedings paper.

Location	Before → after; assumptions and limits
Identification setup	Make explicit positive observational laws, independent graph-permitted roots, complete joint kernels for every clamp value of each supplied target set, and identification uniform over finite observed domains. Negative models may enlarge domains while preserving named values. Conditional cells require positive original evidence.
Theorem 4; Appendix A	The CUT ancestry premise deleted incoming edges only at $\mathbf{Z} \to$ delete them at $\mathbf{X} \cup \mathbf{Z}$. Minimize natural interventions first; retain the iff over every summand with a direct path-recording converse.
Unconditional results; Appendix C	Replace the invalid repeated-response multiplier and marginalization argument by normalized response rows, parent records, and a localized positive gID witness. Apply event equalities before rejecting factors. Preserve the factor/IDENTIFY architecture and unconditional completeness for finite hard/natural syntax. Correct the related procedure and two unconditional example formulas.
Lemma 3; Appendix D	Identification of a specified conditional cell was equated with identification of an enlarged joint quantity → compile numerator and evidence separately and reduce their whole ratio. Connected response-table cumulants give a complete fixed-alphabet test, including positive-evidence countermodels and a denominator valid at every admissible input.
Algorithm 2 and Theorem 6	Replace the conditional completeness argument by a finite algebraic procedure: a full-experiment witness bound, source-intrinsic coordinates, finite alphabet-grid identities, and exact singleton reductions. The original short ancestral-component procedure can give false positives. The new conditional procedure has no efficiency guarantee. The unconditional routine remains a sufficient shortcut.
Conditional running example	The removed T_{r_1} event is dependent on the naturally evaluated T_{r_0} through their common mechanism → retain their joint response factor. For $r_0 \neq r_1$, two positive binary models agree on all experiments and give values $3/4$ and $1/4$. Give the corrected identifiable $r_1 = r_0$ case, restoring normalization and the $R = r_0$ evidence.
Summary, proof details and references	Qualify conditional efficiency; use the published Evans–Richardson (2019) head parameter, intervening only on external parents. Clarify merge induction, event-restricted composition, companion mechanisms, and example conditions; these do not change the reviewed conclusions.

Verification scope. Symbolic AI review covered the changed unconditional and conditional arguments under the stated assumptions, using the cited published results. Earlier fixed counterexample calculations are retained. No new experiment, benchmark, implementation of the algebraic procedure, or formal proof verification was performed for this release.

Cite the original paper; if using this correction, add its unofficial status, date, and URL in a bibliographic note: `https://www.sanghacklee.me/ai-corrected/R79_AI_corrected_2026-10-21.pdf`.

AI review sign-off: claude-fable-5-1 · 2026-10-21.